Contract Requirement: Crane, Multi-Purpose Machine, and Material Handling Equipment (MHE) Operations at Joint Base Pearl Harbor-Hickam (JBPHH)

Encl: (1) NAVFAC HI QA/Safety Contractor & Rigging Crane Entry Package (NAVFAC Hawaii Form 2026)
(2) Figure 10-3 Danger Zone for Cranes and Lifted Loads Operating Near Electrical Power Lines (NAVFAC P-307) dtd Jan 2025

1. Purpose and Scope

All contractors performing work under contracts, task orders, or delivery orders issued by Pearl Harbor Naval Shipyard and IMF (PHNSY&IMF), Code 400, at JBPHH shall comply with the following requirements for the use of:

a. Cranes or
b. Multi-purpose machines, Material Handling Equipment (MHE) or other equipment used in a crane-like application to lift suspended loads.

These requirements are derived from NAVFAC P-307, *Management of Weight Handling Equipment (WHE)*, and other applicable standards.

2. Pre-Operational Submittal and Approval Process

a. **Initial Notification (15+ Working Days Prior)** - The contractor shall notify the Contracting Officer or Contracting Officer's Representative (COR) at least fifteen (15) working days before any crane enters the activity or any MHE is used to lift suspended loads.

b. **Mobile Crane Entry Package (10+ Working Days Prior)** - The contractor shall provide a complete "Contractor Mobile Crane Entry Package" to the COR no less than ten (10) working days prior to the operation. The package must include:

i. Name of contractor and Point of Contact (POC)
ii. Description of contractor's Lifting and Handling (L&H) operation(s)
iii. Date, duration, and location of contractor's L&H operations(s)
iv. WHE or MHE annual and quadrennial inspection records
v. Copies of operator's medical certificate and qualifications by a source that qualifies Crane Operators
vi. Personnel designated and qualified by the Crane Contractor conducting weight handling operations to perform as a rigger shall be qualified in accordance with paragraph g.3.
vii. Copy of the load chart for the specific crane or MHE
viii. Crane Data sheet with but not limited to axle load and axle spacing.
ix. Job site ground loading conditions, outrigger load and cribbing plan.
x. Filled out Enclosure (1) Waterfront Crane/Vehicle Usage Request

xi. Filled out Enclosure (1) Figure P-1 Certificate of Compliance. This involves certifying that the crane, MHE, or rigging equipment meets applicable OSHA and ANSI/ASME regulations, with specific citations for each.

xii. Waterfront operational permit, if applicable – (communicate with COR to obtain permit)

xiii. Critical lift plan(s) when critical lift is to be performed. Refer to 3.d. This plan shall include the following as applicable:

1. The size, weight, and center of gravity of the load to be lifted, including crane (or other machine) equipment and rigging equipment that add to the weight. The OEM's maximum load capacities for the entire range of the lift shall also be provided.

2. The lift geometry, including the crane (or other machine) position and configuration, boom length and angle, height of lift, and radius for the entire range of the lift. This shall include any matting/cribbing and ground preparation, as required. Applies to both single and multiple-crane/machine lifts.

3. A rigging plan, showing the lift points, rigging equipment, and rigging procedures.

4. The environmental conditions under which lift operations are to be stopped.

5. For lifts of personnel, the plan shall demonstrate compliance with the requirements of 29 CFR 1926.1431.

6. For barge mounted mobile cranes, barge stability calculations identifying crane placement/footprint; barge list and trim based on anticipated loading; and load charts based on calculated list and trim specific to the barge the crane is mounted on. The amount of list and trim shall be within the crane OEM's requirements.

7. For lifts in the vicinity of overhead power lines (i.e., if any part of the crane or other machine, including the fully extended boom of a telescoping boom crane or machine, or the load could approach the distances noted in Encl (4) Figure 10-3 Danger Zone for Cranes and Lifted Loads Operating Near Electrical Power Lines during a proposed operation, the plan shall demonstrate compliance with 29 CFR 1926.1408-1411.

c. **Final Entry Request (3+ Working Days Prior)** - Once the Mobile Crane Entry Package is approved, the contractor shall submit the final request for mobile crane entry no less than three (3) working days prior to the desired entry date.

3. **On-Site Operational Requirements**

a. General Safety Compliance - The contractor shall comply with all JBPHH and PHNSY&IMF regulations. The following rules are mandatory:

i. Operator Presence: The operator will not leave the crane cab with a load suspended.

ii. No Side Loading: The crane hook must be positioned directly over the load at all times.

iii. Tail Swing: A physical barricade must be positioned to prevent personnel from entering the crane's tail swing radius.

iv. Stability: "Lift-off" of tires (when on rubber) or outriggers is strictly prohibited.

v. Ground Loading: Adhere to all ground loading limitations. If conditions are questionable, work must stop until the contractor and Contracting Officer mutually determine it is safe to proceed, which may require an engineering study before the crane enters or sets up at the location.

vi. Certification Posting: The approved Certificate of Compliance (Enclosure 1) shall be posted conspicuously in the cab of the crane/vehicle.

vii. Rigging Gear Certification: Provide written certification that all rigging gear meets applicable 29 CFR 1926 regulations.

b. Applicable Industry Standards - The contractor shall comply with all applicable ANSI, ASME, or other consensus standards, including but not limited to: ASME B30.5 (Mobile Cranes), B30.9 (Slings), B30.26 (Rigging Hardware), and ANSI/ITSDF B56.6 (Rough-Terrain Forklifts).

c. Floating and Barge-Mounted Cranes - Floating cranes and barge-mounted mobile cranes require third-party certification from an OSHA-accredited organization (or a third-party certification from a state accredited organization for those states with OSHA approved state plans). They must be equipped with a load indicating device, a wind speed indicator, and a marine list/trim indicator. Operations shall follow the rules of 29 CFR 1926.1437.

d. Critical Lift Requirements - A written Critical Lift Plan is required for any lift meeting the criteria below. The plan must include details on load weight, lift geometry, rigging, and environmental stop-work conditions. A lift is considered CRITICAL if it involves:

i. Lifts over 75% of the capacity of the crane, hoist or other machine (lifts over 50% of the capacity of a barge-mounted crane's hoists) at any radius or lift.

ii. Lifts using more than one crane, hoist, or other machine.

iii. Lifting personnel (lifts of personnel suspended by rigging equipment from multi-purpose machines, MHE, or equipment shall not be permitted).

iv. Operating near overhead power lines (see 29 CFR 1926.1408-1411 and Enclosure 2).

v. Erecting mobile, tower, derrick, or floating cranes.

vi. Lifting submerged or partially submerged objects. The following lifts are not considered critical:

1. Removal of valves, rotors, pipes, etc., from dip tanks for cleaning or coating purposes.

2. Lifting boats of known weight from the water if the boats are of open design with bilge compartments accessible for visual inspection; the boats have label plates indicating weights; and the boats have pre-determined lifting points established by the OEM or the activity engineering organization.

3. Lifting submerged or partially submerged objects that meet the following criteria: the object is verified to not contain fluid in pockets and/or voids that is unaccounted for in the weight of the object; the object is verified or known to not be stuck by suction or adhesion by corrosion, marine growth, excessive surface tension, mud, etc.; and the object is verified to be clear of obstructions, such as other objects in the water, or underwater cables.

vii. Lifts involving binding conditions.

viii. Lifts of hazardous materials. Hazardous materials, e.g., poisons, corrosives, and highly volatile substances. This does not include palletized unit loads of ordnance, nor materials such as oxygen, acetylene, propane, diesel fuel, or gasoline in cans, or tanks that are properly secured in racks or stands designed for lifting and transporting by crane.

ix. Any lift with non-routine rigging, sensitive equipment, or unusual safety risks.

e. Accident and Near Miss Reporting and Investigation

i. Immediate Notification (Within 4 Hours) - The contractor shall notify the Contracting Officer of any WHE accident and Near Miss including Lower Threshold Crane Accidents (LTCAs) as soon as practical, but no later than four (4) hours after the event.

ii. Scene Preservation - The contractor shall secure and preserve the accident scene until released by the Contracting Officer.

iii. Investigation and Corrective Action –The contractor shall investigate to establish the root cause(s) of any WHE accident or Near Miss. However, for a Near Miss, the Contracting Officer may waive the investigation requirement depending on the significance of the event. Except for an LTCA, crane operations shall not proceed until the cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. For an LTCA, the evolution may be allowed to continue, with supervisor authorization and correction of the immediate cause. Upon completion of the job or evolution, the contractor shall stop operations and follow normal accident protocol (i.e., investigation to determine causes and required corrective actions).

iv. Formal Report (Within 30 Days) - The contractor shall provide a formal accident or near-miss report to the Contracting Officer within 30 days using the appropriate form from NAVFAC P-307, Section 12.

v. See Section 4 below for official definitions of "Crane Accident," "LTCA," "Near Miss," and "Rigging Accident."

f. Personnel Qualifications - The contractor shall certify all personnel qualifications via the Certificate of Compliance (Enclosure 1).

i. Crane Operator – Shall be qualified by a recognized source (e.g., union, government agency, or accredited testing organization). Proof of current qualification is mandatory. Additional details regarding crane types and qualifications are provided below.

1. Mobile, Truck-Mounted, and Articulating Boom Cranes: For mobile cranes, commercial truck mounted cranes, and articulating boom cranes with OEM rated capacities of greater than 2000 pounds, the crane operator shall be designated as qualified by a source that qualifies crane operators (i.e., a union, a government agency, or an organization that tests and qualifies crane operators). Proof of current qualification shall be provided.

2. Operator Training and Physical Qualifications: The contractor shall certify via Encl (2) P-1 Certificate of Compliance that the operator is qualified and trained for the operation of the crane or machine to be used, including physical qualifications of the applicable ANSI, ASME, or other appropriate industry/national/international consensus standards (e.g., ASME B30.5 for mobile cranes, ANSI A92.2 for aerial lifts/platforms with hoists or "sign trucks").

ii. Signal Person – Shall be qualified in accordance with 29 CFR 1926.1428 when NAVSEA Standard Item 009-040 is invoked by the contract.

iii. Rigger – Shall be certified as qualified by one of the following methods:

1. A formal apprenticeship program;
2. Completion of training from a recognized source; or
3. A signed statement of compliance.

4. Official Accident Definitions (from NAVFAC P-307) – This section defines the official terminology for accidents and near misses related to weight handling operations, as required for reporting.

a. Clarifications on Weight Handling Equipment Accident and Near Miss Reporting

i. Lower Threshold Crane Accidents (LTCAs):

1. Collision accidents resulting in only superficial cosmetic damage (e.g., scratched paint, scuffs, paint transfer) that would not normally require repair are categorized as LTCAs. Reporting LTCAs helps recognize events at the lowest level.
2. A collision is NOT an LTCA if it causes dents or deformation (regardless of repair needs), or if it involves lifts of high-value/sensitive equipment, molten metals, nuclear materials, or hazardous materials.
3. The LTCA categorization may only be used for ordnance lifts if engineered protection/safety features are present.

b. Dropped Load Exceptions:

i. The fall of minor, loose debris (e.g., paper signs, tie wraps, gravel stuck to a pallet) is not considered a dropped load if there is no potential for injury or damage.

ii. These events may be reported as a Near Miss. Final classification is at the discretion of the Navy Crane Center.

c. Near Misses - A Near Miss is an unplanned event during a weight handling operation that did not result in a definable accident but easily had the potential to do so. A near-miss report is intended to learn from lower-level situations to avert a more severe event. The investigation should be commensurate with the significance of the event.

d. Crane Accidents

i. Operating Envelope - The definition of a crane accident is based on the concept of an "operating envelope." The Operating Envelope: An "operating envelope" is established around any crane operation and consists of the following elements:

1. The crane itself.

2. The operator.
3. All rigging personnel (riggers, signal persons, crane walkers).
4. Other personnel involved (supervisors, mechanics, tagline handlers).
5. The rigging gear between the hook and the load.
6. The load.
7. The crane's supporting structure (ground, rails, etc.).
8. The lift procedure.

ii. Crane Operation: The use of a crane, loaded or unloaded, once an operating envelope is established. This includes setup, breakdown, and repositioning of the crane or hook.

iii. Definition of a Crane Accident: A crane accident occurs when any element within the operating envelope fails to perform correctly during a crane operation, resulting in any of the following:

1. Personnel Injury or Death (Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process of the activity in lieu of these requirements.) Personnel injuries occurring within the operating envelope not directly related to the weight handling operations shall be reported as unplanned occurrences in addition to the normal personnel injury reporting process of the activity.
2. Material or Equipment Damage.
3. Dropped Load (includes any part of the load or rigging gear and any item lifted with the load or rigging gear).*
4. Derailment.*
5. Two-Blocking.*
6. Overload (includes load tests when the nominal test load is exceeded).*
7. Overturned Crane.*
8. Collision (avoidable contact between the load, crane, and/or other objects). Avoidable contact is defined as contact that would have been prevented with proper lift planning and execution. Simply briefing that contact will occur is not sufficient.

Note: Events marked with an asterisk (*) are defined as accidents and must be reported as such, regardless of whether they result in material damage or personal injury."

e. Rigging Accidents

i. Operating Envelope: A rigging accident is defined by its own "operating envelope." The Rigging Operating Envelope: This includes:

1. The rigging gear or miscellaneous equipment itself.
2. The user of the gear (including MHE operators).
3. Other personnel involved in the operation.
4. The load.
5. The gear's supporting structure (padeyes, building structure, etc.).
6. The load's rigging path.
7. The rigging or lift procedure.

ii. Definition of a Rigging Accident: A rigging accident occurs when any element in the rigging operating envelope fails to perform correctly, resulting in any of the following:

1. Personnel Injury or Death. (Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process of the activity in lieu of these requirements.) Personnel injuries occurring within the operating envelope not directly related to the weight handling operations shall be reported as unplanned occurrences in addition to the normal personnel injury reporting process of the activity. Material or Equipment Damage that requires the damaged item to be repaired to perform its intended function This does not include superficial damage such as scratched paint, minor lagging damage, or normal wear on rigging gear.
2. Dropped Load (includes any part of the load or rigging gear).
3. Two-Blocking of cranes and powered hoists.
4. Overload (includes overload of supporting structures and exceeding load test tolerances).

NAVFAC HI QA/Safety Contractor & Rigging Crane Entry Package

1. **Contractor Crane & Rigging Entry Control Package (CCECP) Flow Chart**
2. **Certificate of Compliance P-1** Shall be fully filled out and signed the machine shall be posted on the machine in clear view while in the job site.
3. **Crane Contractor** provides information requested in the CCECP then submits to NAVFAC Contractor Crane Equipment Specialist (CCO) for acceptance prior to allowing the machine/crane entry to the installation.
4. **Certifications - Qualifications** Operator licenses, medical certificates,drivers license (CDL), Rigger & signalman qualification and letter of designations, machine annual certification/inspection, machine/crane (actual load charts) that will be used to load counterweights and the primary lift.
5. **Rigging Equipment Procedures** are the same when rigging gear is used on Multi-Purpose Machines, Fork Lift, Excavator for hoisting. **Company lift plan may be submitted.**
6. **Rigging Diagram Sketching -** showing type of rigging and working load limits for each piece of rigging gear, including sling angle stress. Rigging gear should also show how the rigging gear is connected to the load.
7. **Contractor Crane & Rigging Entry Control Package-** Shall be submitted to the GOV Core Rep & Contractor Crane Equipment Specialist for review, once accepted scheduling date time and location accordingly for Government Oversight.
8. **Hoisting Equipment Activity Hazard Analysis (AHA)** – A detailed AHA will be completed by the contractor and submitted to NAVFAC prior to the start of work. The AHA will be briefed at the job site with the GOV Rep. or alternate CCO & reviewed at the Preparatory/Initial Control Phase Meetings.
9. **Crane & Rigging Gear Accident Report/Instructions** – To be completed by contractor when a WHE mishap occurs in accordance with NAVFACP-307 Section 12
10. **Sample Rigging Diagram/Sketch** – Constructed by contractor on Pre-Entry Checklist or as an attachment. - AHA filled out with required safety and controls listed.
11. Contractor familiarization with NAVFAC P-307 Section 11, 7, 8, 9, 10, 12, 14 recommended.

APPENDIX P – CONTRACTOR CRANE ALTERNATE MACHINE & RIGGING OPERATION USED TO LIFT SUSPENDED LOAD REQUIREMENTS

FIGURE P-1

CERTIFICATE OF COMPLIANCE

This certificate shall be signed by an official of the company that provides cranes (or multi-purpose machines, MHE, or construction equipment used to lift loads suspended by rigging gear) or rigging gear for any application under this contract. Post a completed certificate on each crane or alternate machine (or in the contractor's on-site office for rigging operations) brought onto Navy property.

CONTRACTING OFFICER'S POC & UIC # (*Gov. Contract Official Representative UIC)*	PHONE
PRIME CONTRACTOR/PHONE	CONTRACT NUMBER
CRANE SUBCONTRACTOR and MACHINE SUPPLIER/PHONE (subcontractor company)	CRANE or machine Serial Number
CRANE OR ALTERNATE MACHINE MANUFACTURER/TYPE/CAPACITY	
MACHINE OPERATOR / RIGGER NAME(S)	

I certify that

1. The above noted crane or alternate machine and all rigging gear conform to applicable OSHA regulations (host nation regulations for naval activities in foreign countries) and applicable ASME B30 or other standards. The following OSHA regulations and ASME or other standards Apply: <u>Applicable 29 CFR parts 1910,1915,1917,1919,or 1926/ASME B30.2/B56.6/B30.9/B30.10/B30.26/ANSI B30.5/EM385-1-1/PCSA #2 and NAVFAC P-307/P300</u>
2. The operators noted above have been trained and are qualified for the operation of the above noted crane(s) or alternate machine(s).
3. All safety devices and operator aids are enabled and functioning properly and the operators noted above have been trained not to bypass safety devices and operator aids during lifting operations.
4. The operators, riggers and company officials are aware of the actions required in the event of an accident as specified in the contract.
5. Signal persons used in construction work are qualified in accordance 29 CFR 1926.1428
6. Riggers are qualified in accordance with NAVFAC P-307, paragraph 11.1.k.
7. All personnel working on the job site have been trained to not stand under a load or in the fall zone of a suspended load unless specifically allowed by USACE EM 385-1-1.

COMPANY OFFICIAL SIGNATURE	DATE
COMPANY OFFICIAL NAME/TITLE	

POST ON CRANE (OR ALTERNATE MACHINE)

(IN CAB OR VEHICLE)

(or in the contractor's on-site office for rigging operations)

APPENDIX P – CONTRACTOR CRANE ALTERNATE MACHINE & RIGGING OPERATION USED TO LIFT SUSPENDED LOAD REQUIREMENTS

FIGURE P-1

CERTIFICATE OF COMPLIANCE

This certificate shall be signed by an official of the company that provides cranes (or multi-purpose machines, MHE, or construction equipment used to lift loads suspended by rigging gear) or rigging gear for any application under this contract. Post a completed certificate on each crane or alternate machine (or in the contractor's on-site office for rigging operations) brought onto Navy property.

CONTRACTING OFFICER'S POC & UIC # (*Gov. Contract Official Representative UIC)*	PHONE
PRIME CONTRACTOR/PHONE	CONTRACT NUMBER
CRANE SUBCONTRACTOR and MACHINE SUPPLIER/PHONE (subcontractor company)	CRANE or machine Serial Number
CRANE OR ALTERNATE MACHINE MANUFACTURER/TYPE/CAPACITY	
MACHINE OPERATOR / RIGGER NAME(S)	

I certify that

1. The above noted crane or alternate machine and all rigging gear conform to applicable OSHA regulations (host nation regulations for naval activities in foreign countries) and applicable ASME B30 or other standards. The following OSHA regulations and ASME or other standards Apply: Applicable 29 CFR parts 1910,1915,1917,1919,or 1926/ASME B30.2/B56.6/B30.9/B30.10/B30.26/ANSI B30.5/EM385-1-1/PCSA #2 and NAVFAC P-307/P300
2. The operators noted above have been trained and are qualified for the operation of the above noted crane(s) or alternate machine(s).
3. All safety devices and operator aids are enabled and functioning properly and the operators noted above have been trained not to bypass safety devices and operator aids during lifting operations.
4. The operators, riggers and company officials are aware of the actions required in the event of an accident as specified in the contract.
5. Signal persons used in construction work are qualified in accordance 29 CFR 1926.1428
6. Riggers are qualified in accordance with NAVFAC P-307, paragraph 11.1.k.
7. All personnel working on the job site have been trained to not stand under a load or in the fall zone of a suspended load unless specifically allowed by USACE EM 385-1-1.

COMPANY OFFICIAL SIGNATURE	DATE
COMPANY OFFICIAL NAME/TITLE	

POST ON CRANE (OR ALTERNATE MACHINE)

(IN CAB OR VEHICLE)

(or in the contractor's on-site office for rigging operations)

Company Name and the designated qualified Crane Rigger / Signalers P-1 Contniuation.

CONTRACTOR CRANE PRE-ENTRY CHECKLIST

1	Crane Company:	Estimated Date of Entry:
	Crane Type/Manufacturer/Model/Serial Number/Crane Number:	Estimated Time of Entry: Government Contracting
2	Date of Annual Inspection Expiration **(attach copy)**	**Expiration Date:**
3	Job Site/ Facility POC Name & phone number **Official** (local representative and UIC code)	**Name:**
		UIC Code & Phone #:

Prepared by: **Date:** **NAVFAC CCO: WES N**

GENERAL INFORMATION

4	Narrative of crane / equipment lifts - Planed lift of the day, heaviest load possible, including location, Crane Counterweight installation con-fig. identified on next page CTW #1, type of standard or rigid dunnage (size for heavy lifts). `		
5	Duration crane will be on the job site (i.e.. four days, two weeks)		
6	Include a valid Certificate of Compliance (P-1) to identify Competent / Qualified Riggers,		
7	What is the weight of the heaviest load to be lifted?		lbs.
8	What is the weight of the rigging gear? **(Please include WLL of slings, shackles and attachments in sketch)**		lbs.
9	**DEDUCTIONS**: For telescopic forklift and excavator, use block (g) for the weight of the forklift attachment or the weight of the excavator bucket. Blocks (a) thru (f) do not apply. **Please be reminded that the lift are limited by the lowest capacity (If hoist line, or hook capacity is the lesser value, that is the Gross Capacity to be used)** Parts Line Used: _________parts (Line Pull Each: __________lbs.) X (No. of Parts line) = 0__________Hoist Line Capacity TOTAL COUNTERWEIGHT: ____________________	**a. Main Block**	lbs.
		b. Aux. Block	lbs.
		c. Jib (Stowed)	lbs.
		d. Jib (Erected)	lbs.
		e. Whip Ball	lbs.
		f. Excess Wire Rope	lbs.
		g. Other	lbs.
		TOTAL DEDUCTIONS (a-f)	0 lbs.
10	**What is the total weight of this lift? (7+8+Total Deductions)**	**TOTAL**	0 lbs.
11	What is the **Gross Capacity, based off line pull or radius? (whichever is less)**		lbs.
12	What **Percentage** of crane capacity is this lift? **Total** × 100 = 0 ÷ **Gross Cap** =		NaN %

		Boom Lenght	JIB	OFFSET
13	What is the **main boom length** planned for the lift? (Item 10 & 11) If Jib is used, indicate the length and offset. For telescopic forklifts list the maximum length of the boom used for the lift.			

14	What is the **maximum radius for the planned** lift? (Item 10 & 11)	**Radius**		**Angle**	

		Boom Length	MAX Radius	Load Weight	Rigging Gear	Main Block	Aux Block	Jib Stowed	Jib Erected	Whip	Wire Rope	Other	Total	Gross Cap	%
15	**CTW#1**												0		NaN
	Lift#2												0		NaN
	Lift#3												0		NaN
	Lift#4												0		NaN

16	If Jib is used, has crane been load tested in that configuration? **(attach copy)**
17	Include a copy of the cranes Range Diagram for the planned boom configuration. **(attach copy)**
18	Include a copy of the Load Chart **Page** (e.g. page 26) for the planned boom configuration. **(attach copy)**
19	Complete the rigging sketch **(item 31)** for all lifts. (please include chaffing gear points in sketch) **(company form is acceptable)**
20	Complete job site sketch **(item 34)**, include underground utilities or overhead power lines. **(company form is acceptable)**
21	Include a valid Operator's Certification, CDL, DOT Medical Certificate by a registered physician (N/A for forklift and excavator operators), Rigger qualifications and a Certificate of Compliance (P-1.) with their names.
22	Include proof of rigger and signalman qualifications and a copy of Photo I.D. **(attach copy)**

		YES	NO	N/A
23	Does the crane have a **wind speed indicating device? (EM385, 16.A.08.l)**	☐	✔	☐
24	Does the crane have a functioning **anti-two block system? (EM385-1-1, 16.E.03)**	☐	☐	
25	Has Airfield/FAA Clearance been obtained?	☐	☐	
26	For crawler crane, does the plan indicate area restrictions for operation?	✔	✔	☐
27	For mobile crane mounted on barge, is crane equipped with load indicating device, wind indicating device, marine type 'list and trim' indicator (in 1/2 degree increments)? Does plan include a revised load chart?	☐	✔	☐
28	For floating crane, does plan include maximum allowable list?	☐	☐	☐
29	Is fall protection required for any part of the job?	☐	☐	

CRITICAL LIFTS

30	NAVFAC P307 Complex lifts include (a) hazardous materials (b) large and complex geometric shapes (c) lifts of personnel (d) lifts exceeding 75% of the crane's main or whip hoist at any radius / 50 percent for barge mounted or mobile cranes gross capacity (e) lifts of submerged or partially submerged objects (f) multiple crane or multiple hook lifts on the same crane (g) lifts of unusually expensive or one-of-a-kind equipment or components (h) lifts of constrained or potentially constrained loads (binding conditions) (i) lifts made in the vicinity of overhead power lines (j) load test. NAVFAC P-307 Section 10 Critical/Complex lifts \.

		YES	NO
31	**Is this a critical lift?** If **Yes**, complete Items 32-37 If not do not fill in items 31 -37	☐	☐

32	Critical lift briefing sign off sheet will be completed on the job site.		
33	Please describe circumstances that classify this operation as a critical lift. (**Refer to NAVFAC P-307**)		
34	Include in rigging/job site sketch the exact dimensions of load and the height to be lifted or lowered. **(company form is acceptable)**		
35	Include boom angle for complex/critical lifts	**Boom Angle**	
36	Describe the the wind speed & ground conditions.		
37	What type of communication will be used? **hand** ☐ **voice** ☐ **radio** ☐ **signals will be relayed** ☐ Fill this in prior to a critical lift		

Print and Sign

Operator____

Rigger in Charge____________________________Date___________

Signalman____

Lift Supervisor____

Other____

Other____

Other____

Other____

Other____

Other____

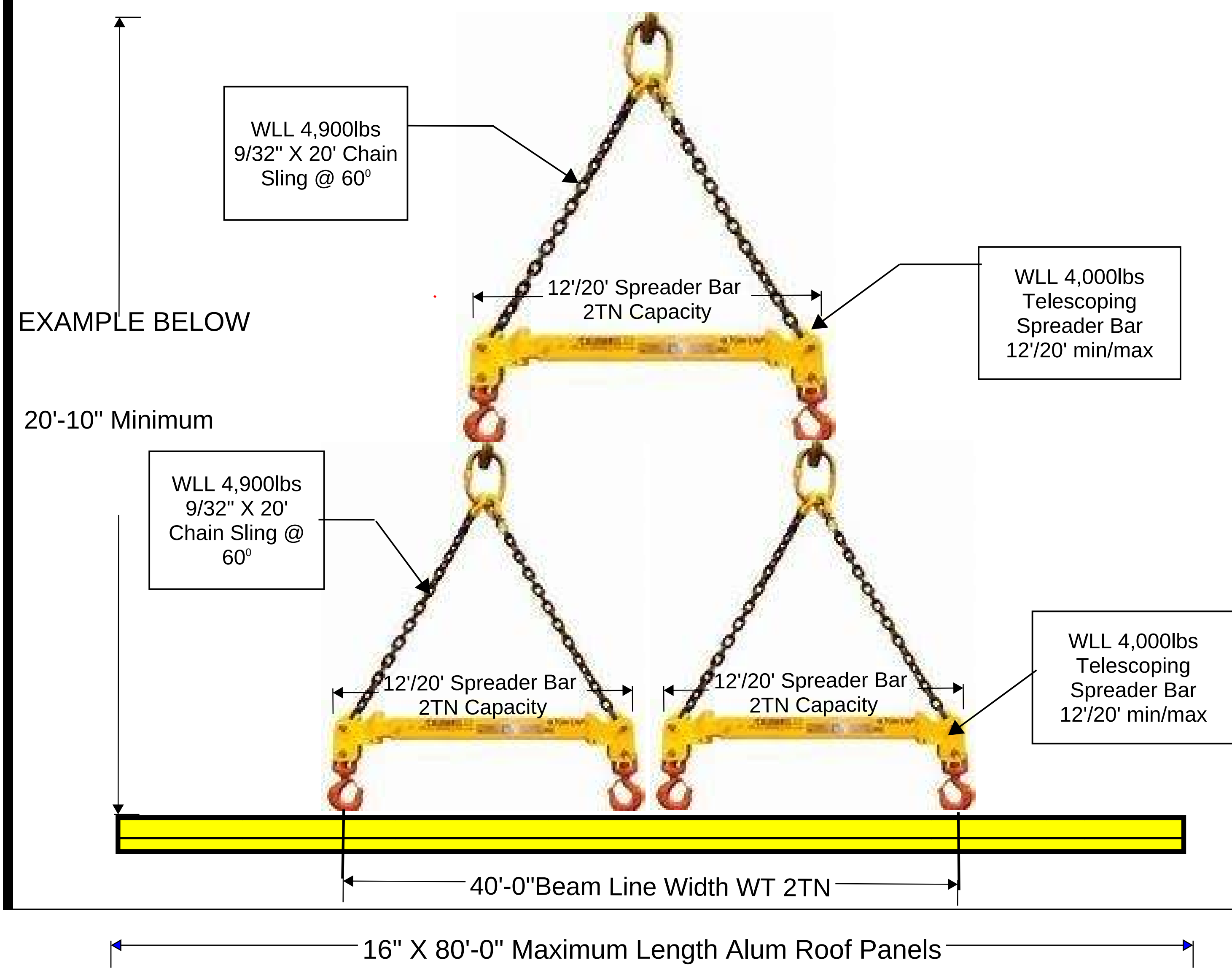
RIGGING DIAGRAM/SKETCH (Draw or Attach)	
34	Construct a rigging diagram below to show the lift points, description of the rigging procedures and hardware requirements (i.e. hooks, shackles, slings, eye bolts, spreader bars) including Working Load Limits (WLL) and angles of any applied slings etc. **See Enclosure 7 for sample:**

EXAMPLE BELOW

RIGGING DIAGRAM/SKETCH (Draw or Attach)	
34	Construct rigging diagram below to show the lift points, description of the rigging procedures and hardware requirements (i.e. hooks, shackles, slings, eyebolts, spreader bars) including Working Load Limits (WLL) and angles of any applied slings etc. **See Enclosure 7 for sample:**

DE-BUILDING 2 REROOF JBPHH HI

RIGGING/DIAGRAM SKETCH



35 **Jobsite Sketch:** (Draw in location of utilities and their proximity to construction site, include such things as; proposed excavations, location of heavy equipment, scaffolding, material storage areas, etc.)

EXAMPLE BELOW


Housings
#5
#12
Kiln
Google Earth
Image Landsat / Copernicus
N
100 ft

Instructions for completing Contractor Activity Hazard Analysis

1. **Activity/Work Task – Insert work/task this AHA is written for i.e. excavation, scaffold building, foundation preparation.**
2. **Enter Project Location**
3. **Enter Contract Number**
4. **Enter Date Prepared**
5. **Enter Prime Contractors name**
6. **Enter Sub Contractors name**
7. **Enter name and title of person who prepared the AHA**
8. **Enter name and title of person who reviewed AHA**
9. **Job steps is the complete sequence of work, not general statements to complete the entire activity**
10. **Hazards is the known safety risks associated with completing the task**
11. **Controls is the safety measures in place to reduce the hazard to the lowest level possible**
12. **Risk Assessment code is where Severity and Probability intersect, place that letter E, H, M, or L in the RAC column**
13. **List all equipment to be used to complete this activity i.e. crane, backhoe, vehicle, all heavy equipment**
14. **List the training requirements required by EM 385, Safety Spec 01356 or OSHA that apply to this task.**
    - **List competent person(s) required for specific tasks in EM 385**
    - **List qualified person(s) required for specific tasks in EM 385**
    - **List CPR/First Aid training and qualification dates**
15. **List all inspection requirements of EM 385, Governmental Safety Requirements Specifications or OSHA 29 CFR 1926**

IAW EM 385 01.A.13 Contractor-Required AHA "Work will not begin until the AHA for the work activity has been accepted by the GDA"
The AHA shall be reviewed and modified as necessary to address changing site condition, operations or change of competent/qualified person's

EXAMPLE BELOW

Activity Hazard Analysis (AHA)

Activity/Work Task:	Overall Risk Assessment Code (RAC) (Use highest code)					RAC Code L
Project Location:	**Risk Assessment Code (RAC) Matrix**					
Contract Number:	**Severity**	**Probability**				
Date Prepared:		Frequent	Likely	Occasional	Seldom	Unlikely
Prepared by (Name/Title):	Catastrophic	E	E	H	H	M
	Critical	E	H	H	M	L
Reviewed by (Name/Title):	Marginal	H	M	M	L	L
	Negligible	M	L	L	L	L

Notes: (Field Notes, Review Comments, etc.)

Step 1: Review each **"Hazard"** with identified safety **"Controls"** and determine RAC (See above)

"Probability" is the likelihood to cause an incident, near miss, or accident and identified as: Frequent, Likely, Occasional, Seldom or Unlikely.	**RAC Chart**
"Severity" is the outcome/degree if an incident, near miss, or accident did occur and identified as: Catastrophic, Critical, Marginal, or Negligible	E = Extremely High Risk H = High Risk
Step 2: Identify the RAC (Probability/Severity) as E, H, M, or L for each "Hazard" on AHA. Annotate the overall highest RAC at the top of AHA.	M = Moderate Risk L = Low Risk

Job	Hazards	Controls	RAC
1. Driving to Site.	Vehicular & pedestrian accidents	Crane & Forklift driver will use defensive driving techniques; move only with proper escorts and permits. Proper PPE shall be worn at all times to include: Hard Hats, construction gloves, hard toe safety shoes, bright colored clothing/safety vest, and personnel flotation device if working at edge of piers or at edge of afloat units.	L
2. Load Handling Equipment Set-up	Equipment & pedestrian accidents	Establish crane zone that will keep non-essential personnel from WHE area. Move equipment only when directed and with line of sight between operator and RIC/TC man. Speeds on piers are no more than 5 MPH. Do not drive wheels over manholes.	L
	Prevent unintended LHE rolling.	Parking brake shall be set whenever WHE is parked. The wheels shall be chocked if parked on an incline.	L
	Personnel protection from rotating structures of the LHE and fixed objects.	The swing radius of the cranes superstructure & counterweights shall be barricaded to prevent personnel from being struck or crushed. Utilize cones and caution tape to keep personnel out of crane zone.	L
	Damaging underground utilities.	If required, verify from NAVFAC FEAD to ensure no underground voids, pipes, boxes, etc. prior to the operation. Cranes must use 6' x 4' steel dunnage under each outrigger. 6' x 4' rigid steel dunnage is required on JBPHH/PHNSY piers.	L

Activity Hazard Analysis (AHA)

Job	Hazards	Controls	RAC
1. Rigging gear installation	Qualified Riggers shall be used. Load falling, damaged rigging gear.	Qualified Riggers inspect rigging gear for any damage, ensure rigging gear has enough capacity to lift load, and ensure rigging gear is properly marked. Ref: ASME B30 and NAVFAC P-307 section 14.	L
2. Install / Remove Counterweights IF Applicable	Qualified Riggers shall rig and derig counterweights. Personnel injuries and equipment damage	Counterweight Installation / removal to be performed in accordance with OEM. All personnel associated with installation of counterweights shall be knowledgeable in manufacturer's procedures and requirements. All personnel shall take special precautions to avoid pinch points and warn others of same. Operator shall only move crane and load when line of site with riggers and / or signalman is maintained. All personnel involved with installing / removing counterweights on the crane deck shall be trained in use of & shall use Fall Protection per 1926.1423(f) to prevent falls, including use of manufacturer's access ladders, walkways & handrails.	L

Job	H	Controls	RAC
3. Lifting loads	Qualified Riggers Qualified Crane Signalers shall Install Rigging and Perform Signaling. Crane winch or brake fails to lift or hold load.	Operator shall engage hoist or activate hoist only when directed by rigger or signalman. Signalman shall maintain clear view of all riggers / installers. Hoisting to be performed within Safe Work Zone. No personnel under load while being lifting or suspended. CRANES: Minimum part hoist line as required by Manufacturer. Operator shall engage hoist only when directed by rigger / signalman. Operator and /or signalmen shall maintain clear view of all riggers / installers. After initial lift the operator shall immediately stop the lift to insure that rigging is set and hoist winch & brake can hold the Load, i. e. **Test Lift – stop the hoist on every lift after the load is off the surface!**	L L

Activity Hazard Analysis (AHA)

Job Steps	Hazard	Controls	RAC
4. Lifting loads	Alert site personnel of lift.	Sound Horn or whistle to notify personnel in area prior to lift. Prohibit access to only essential personnel.	L
	Operator unable to see loads & or Riggers /Installers for duration of lifts.	Signalman shall direct load in conjunction with riggers / installers and maintaining constant contact with operator with use of hand signals or radios equipped with unique channel, throughout operations. Operator or signalman shall maintain clear view of all riggers / installers. Only standard hand and voice signals shall be used. Operator shall stop all operations should communication with signal persons fail. Anybody can call safety time out.	L

Job Steps	Hazard	Controls	RAC
5. Lifting loads	Sling or Rigging Failure	Operator & Rigger shall ensure proper Sling / rigging selection and condition. Daily Inspections or more frequent if indicated. Only rigging equipment that has affixed legible identification markings that indicates the safe working load will be used. Use sling softeners for areas that can damage slings.	L
	Two-Blocking of Hook block or Ball & Boom top. IF Applicable to Equipment	Use of anti-two block device: Operator shall not over-ride nor disable. Operator shall stop lift if device is activated. If not available, operator & signalman shall maintain line of sight with each other and the two-block area.	L

Activity Hazard Analysis (AHA)

Job Steps	Hazards	Controls	RAC
6. Lifting loads	Overhead power lines:	Site Inspection revealed no OH power lines within a boom length of this operation. Controlling Entity shall alert all personnel of the position of the OH Lines. See 29 CFR 1926.1408, EM 385-1-1-1 sec 16-1, NAFVAC P-307 sec 10.	L
	Wind Considerations:	Operator shall remain vigilant to riggers' loss of control of load. At wind speeds measured by a wind indicating device of 20 mph or greater all hoisting operations shall cease. Operator, Rigger & Lift Supervisor shall consult & determine if operations can safely proceed. If winds are at or predicted to exceed 30 MPH: all operations shall cease & the crane shall be secured from operations. Boom mounted Anemometer shall be used to determine wind speeds.	L

Job Steps	Hazards	Controls	RAC

Activity Hazard Analysis (AHA)

7. Lifting loads	Spinning or load control	Use Tagline(s) when needed to help control loads.	**L**
	Lightning	When conditions are such that lightning is observed: all crane & hoisting operations shall cease. A period of 30 minutes between subsequent observations shall be observed prior to resuming work.	**L**
	Crushing fingers, hands during assembly	Signal person shall take added precaution to insure all riggers and / or installers are clear of any potential load drop. Blocking & shims should be used to prevent crushing.	**L**

Activity Hazard Analysis (AHA)

8. Lifting and slewing loads near personnel. Head, Hand, Foot and Eye Injury's	Under the fall zone / getting under the load Personnel Protection Equipment (PPE)	All personnel must stay clear of the fall zone unless directed by the rigger in charge. Stop load movement first. Personnel are never allowed under the load **handle loads at chest height not above the head .** Personnel Are Never Allowed Under The Load: Ref NAVFAC P-307, EM 3851-1, 29 CFR. Establish crane zone (barricade) by using cones, caution or danger tape, designate safety personnel if required to keep personnel away of crane operations. PPE must be worn at all times including: Bright vest, gloves, hard toed shoes, safety glasses, hard helmet.	**L**
Crane / Equipment MHE Accident	Dropping of load, banging load against object, damaged rigging gear, Vehicle/ Machine Accident/ Damage Personnel injury.	Stop evolution, make operation safe, call 911 if necessary, notify Contracting Officer and Contractor Crane Oversight (CCO) personnel. Wait for further direction by Contracting Officer and CCO. Keep the situation in lock down for photos. Submit crane accident or near miss report	

LHE Operational Requirements ï Fueling (NA for PRL-PND-611)	• Fire from static Discharge or ignition of vapors	• LHE shall be shut down before and during fueling operations. Closed systems, with an automatic shut-off that will prevent spillage if connections are broken, may be used to fuel diesel powered equipment left running. • One (minimum) dry chemical or CO2 fire extinguisher with a minimum rating of 10-B:C installed in the cab or at the machinery housing	**L**
(Space for field personnel to write-in additional site-specific steps, hazards, and controls)	•	•	

Equipment	PPE	Competent or Qualified Personnel Name(s)	Inspection	Training
• Personal Vehicles • Fire Extinguishers • First Aid Kits • Power Tools • Small hand tools • Heavy Trucks • LHE • Rigging • Cribbing • Engine Hoist	• Hard hats • Safety toed boots with ankle supports • Safety glasses • Safety vests (Class 2 Reflective Safety Vest) • Gloves (as needed) • Insulated gloves (as needed) • Hearing protection (as needed)	• Competent Person: • • Qualified Person: **D** •	• Initial inspection of all equipment and tools to be used. • Daily inspection of equipment and tools. • Daily EHS documented inspections. • PPE inspections.	• NAVFAC P-307 • First aid/CPR/BBP • Initial site safety briefing • Daily tailgate safety briefings • Vehicle training • Proper license to operate heavy equipment • Fall Protection (potential for assembly of LHE) • Medical Clearance (if required)

AHA Signature Page

In accordance with NAVFAC Supervision "Work will not begin until the AHA for the work activity has been completed. The AHA shall be reviewed and modified as necessary to address changing site conditions, operations or change of competent/qualified persons. By signing below I understand, agree to, and will conform to the site rules set forth in this AHA, my respective company's Planning Documents (including amendments and attachments), and those controls agreed upon during any site-specific health and safety briefing(s).			
Name	**Signature**	**Date**	**Briefing By**

Equipment to be Used	Training Requirements/Competent or Qualified Personnel name(s)	Inspection Requirements
Load Handling Equipment: Crane Multipurpose Machine Forklift Hoists & Rigging	NCCCO or Equivalent Certification for Operator provided under separate cover for: or alternate. NCCCO or Equivalent Certified Rigger / Crane signaler Signal:	Operator & Crane Certification CDL License Medical Certification Operators Daily Crane Inspection Document Rigger / Crane Signaler Qualifications

:

Prepared by (Print Name, Sign and Date):

Crew Supervisor Review (Print Name, Sign and Date): ______________________________

AHA Discussed with Crew at Preparatory Meeting Held On: ___________

Sign:___________________________ Sign: ___________________________ Sign: ___________________________

Print:___________________________ Print: ___________________________ Print: ___________________________

Sign:___________________________ Sign: ___________________________ Sign: ___________________________

Print:___________________________ Print: ___________________________ Print: ___________________________

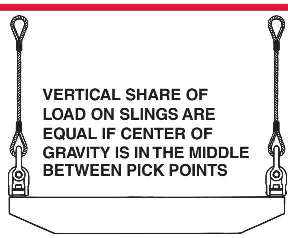
Sign:___________________________ Sign: ___________________________ Sign: ___________________________

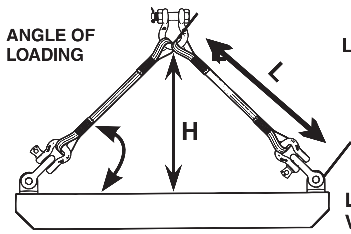
Print:___________________________ Print: ___________________________ Print: ___________________________

Add additional pages as needed for additional signatures.

SLING ANGLES

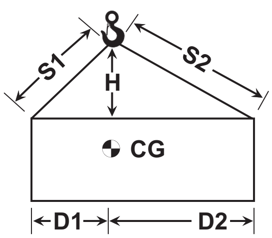
TWO LEGGED SLING - WIRE ROPE, CHAIN, SYNTHETICS





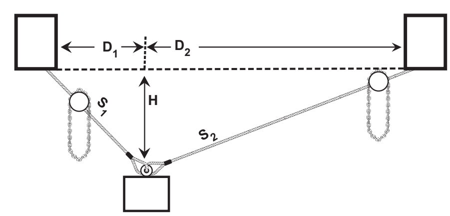
ANGLE OF LOADING (A) DEGREE	LOAD ANGLE FACTOR = L/H
90	1.000
60	1.155
50	1.305
45	1.414
30	2.000

LOAD ON EACH LEG OF SLING =
VERTICAL SHARE OF LOAD X LOAD ANGLE FACTOR



LOAD ON SLING CALCULATED
TENSION 1 = LOAD X D2 X S1/(H(D1+D2))
TENSION 2 = LOAD X D1 X S2/(H(D1+D2))

ANGLE OF LOADING OF LESS THAN 30 DEGREES ARE NOT RECOMMENDED REFER TO ASME B30.9 FOR FULL INFORMATION



LOAD ON SLING CALCULATED
TENSION 1 = LOAD X D2 X S1/(H(D1+D2))
TENSION 2 = LOAD X D1 X S2/(H(D1+D2))

OPERATING PRACTICES - ASME B30.9

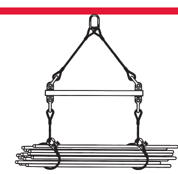
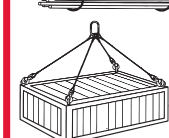
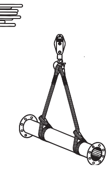
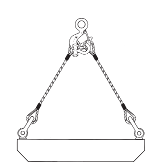
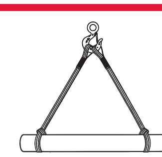
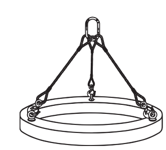
WHENEVER ANY SLING IS USED, THE FOLLOWING PRACTICES SHALL BE OBSERVED.

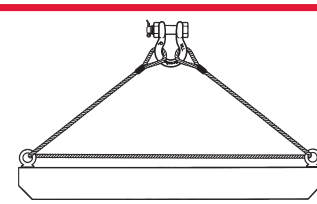
1. SLINGS THAT ARE DAMAGED OR DEFECTIVE SHALL NOT BE USED.
2. SLINGS SHALL NOT BE SHORTENED OR LENGTHENED BY KNOTTING OR TWISTING.
3. SLING LEGS SHALL NOT BE KINKED.
4. THE RATED LOAD OF THE SLING SHALL NOT BE EXCEEDED.
5. SLINGS USED IN A BASKET HITCH SHALL HAVE THE LOADS BALANCED TO PREVENT SLIPPAGE.
6. SLINGS SHALL BE SECURELY ATTACHED TO THEIR LOAD.
7. SLINGS SHALL BE PROTECTED FROM EDGES, CORNERS, PROTRUSIONS AND ABRASIVE SURFACES TO PREVENT SLING DAMAGE.
8. DURING LIFTING, WITH OR WITHOUT LOAD, PERSONNEL SHALL BE ALERT FOR POSSIBLE SNAGGING.
9. ALL EMPLOYEES SHALL BE KEPT CLEAR OF LOADS ABOUT TO BE LIFTED AND OR SUSPENDED LOADS.
10. HANDS OR FINGERS SHALL NOT BE PLACED BETWEEN THE SLING AND ITS LOAD WHILE THE SLING IS BEING TIGHTENED AROUND THE LOAD.
11. SHOCK LOADING SHOULD BE AVOIDED.
12. A SLING SHALL NOT BE PULLED FROM UNDER A LOAD WHEN THE LOAD IS RESTING ON THE SLING.

INSPECTION: EACH DAY BEFORE BEING USED, THE SLING AND ALL FASTENINGS AND ATTACHMENTS SHALL BE INSPECTED FOR DAMAGE OR DEFECTS BY A COMPETENT PERSON DESIGNATED BY THE EMPLOYER. ADDITIONAL INSPECTIONS SHALL BE PERFORMED DURING SLING USE WHERE SERVICE CONDITIONS WARRANT. DAMAGED OR DEFECTIVE SLINGS SHALL BE IMMEDIATELY REMOVED FROM SERVICE.

LOAD CONTROL

POSITIVE LOAD CONTROL



REEVING THROUGH CONNECTIONS TO LOAD INCREASES LOAD ON CONNECTION FITTINGS BY AS MUCH AS TWICE.

DO NOT REEVE!

TUFLEX ENDLESS ROUNDSLINGS

Tuflex Endless (EN)
The Most Versatile *Tuflex* Roundsling

Features, Advantages and Benefits

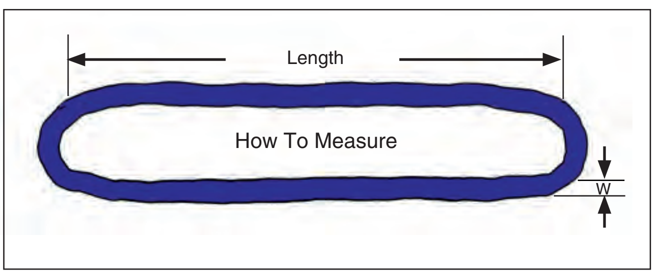
Maintains all the basic *Tuflex* features plus...

Promotes Safety

- Load stability and balance can be achieved by spreading sling legs.

Saves Money

- Wear points can be shifted to extend sling life
- The most flexible style of sling



Part No.	Color		Rated Capacity (lbs.)* Vertical	Choker	Basket @ 90°	Basket @ 45°	Minimum Length (ft.)	Approximate Measurements Weight (lbs. / ft.)	Body Dia. Relaxed (in.)	(W) Width at Load (in.)	Minimum Hardware Dia. ** (in.)
EN30	Purple		2,600	2,100	5,200	3,600	1 1/2	.2	5/8	1	7/16
EN60	Green		5,300	4,200	10,600	7,400	1 1/2	.3	7/8	1 3/8	5/8
EN90	Yellow		8,400	6,700	16,800	11,800	3	.5	1 1/8	1 3/4	3/4
EN120	Tan		10,600	8,500	21,200	14,000	3	.6	1 1/8	1 7/8	7/8
EN150	Red		13,200	10,600	26,400	18,000	3	.8	1 3/8	2	1
EN180	White		16,800	13,400	33,600	23,000	3	.9	1 3/8	2 1/8	1 1/8
EN240	Blue		21,200	17,000	42,400	29,000	3	1.3	1 3/4	2 5/8	1 3/16
EN360	Grey		31,000	24,800	62,000	43,000	3	1.7	2 1/4	3 1/4	1 1/2
EN600	Brown		53,000	42,400	106,000	74,000	8	2.8	2 3/4	4	2
EN800	Olive		66,000	52,800	132,000	93,000	8	3.4	3 1/8	4 5/8	2 1/8
EN1000	Black		90,000	72,000	180,000	127,000	8	4.3	3 5/8	5 1/4	2 1/2

* **WARNING** Do not exceed rated capacities. Sling capacity decreases as the angle from horizontal decreases. Slings should not be used at angles of less than 30°.
Refer to Effect of Angle chart page 12.
** This is the smallest recommended connection hardware diameter to be used for a vertical hitch.

CRANE AND RIGGING NEAR MISS REPORT

Near Miss Category: ☐ Crane Near Miss ☐ Rigging Near Miss

Name and UIC of Reporting Activity:			**Copy To: Navy Crane Center** **Bldg. 491 NNSY** **Portsmouth, VA 23709** **Fax: 757-967-3808**	
Name and UIC of Activity Responsible for the Near Miss:	**Activity Name and UIC of Event Location:**		**Report No:**	
	Specific Location:	**Date:**	**Time:**	
BOS Contractor: ☐ Yes ☐ No	**Other Contractor** ☐ Yes ☐ No If Yes, Contract #________________________			
Crane No:	**Crane Type:**	**Category:**	**Crane OEM:**	
Crane Capacity:	**Hoist Capacity:**	**Weight of Load on hook:**	**Weather:**	
Critical Lift or Critical Non-Crane Rigging Operation? ☐ Yes ☐ No				
Is this near miss indicative of a recurring problem? ☐ Yes ☐ No				

In the space below, include a description of the event, root cause and corrective actions taken to prevent recurrence:

Brief description:

Root cause:

Corrective Actions:

INCLUDE: Printed Name, Code and Date unless otherwise specified.

Preparer:	**Phone:**	**E-mail:**	**Code:**	**Date:**
Approved by:	**Phone:**	**E-mail:**	**Code:**	**Date:**

Figure 12-2 (1 of 2)

CRANE AND RIGGING NEAR MISS REPORT INSTRUCTIONS

1. Near Miss Category: Indicate crane or rigging near miss.

2. Reporting Activity/UIC: The activity and unit identification code responsible for reporting the near miss in accordance with the guidance of paragraph 12.6.2.

3. Activity Responsible for the Near Miss/UIC: Same as #2 above, or for NAVFACENGSYSCOMs, provide the FEC level UIC.

4. Report No.: The activity assigned near miss number (e.g., Activity UIC-FY-CA-01).

5. Location UIC: The activity and unit identification code where the event took place.

6. Specific Location: The detailed location where the near miss took place (e.g., building 213, drydock 5).

7. Near Miss Date: The date the near miss occurred.

8. Time: The time (24-hour clock) the near miss occurred (e.g., 1300).

9. Is the responsible party a BOS Contractor? Check yes or no.

10. Is the responsible party a contractor other than a BOS contractor? Check yes or no. If yes, enter contract number.

11. Crane No.: The activity assigned crane number (e.g., PC-5), if applicable.

12. Crane Type: The type of crane involved in the near miss (e.g., mobile, bridge), if applicable.

13. Category: Identify category of crane (i.e., 1, 2, or 3), if applicable.

14. Crane OEM: The original equipment manufacturer of the crane (e.g., Samsung, Grove, P&H), if applicable.

15. Crane Capacity: The certified capacity of the crane (e.g., 120,000 pounds), if applicable.

16. Hoist Capacity: The certified capacity of the hoist involved in the near miss at the max radius of the operation, if applicable.

17. Weight of Load on Hook: If applicable, the weight of the load on the hook.

18. Weather: The weather conditions at time of accident (e.g., wind, rain, cold).

19. Critical lift: Was the crane or rigging gear being used in a critical lift?

20. Is this a recurring problem? Check yes or no. Identify any other similar near misses or accidents.

21. Brief Description: No more than one paragraph summarizing the resultant incident.

22. Root Cause and Detailed Description: Provide the relevant background in a descriptive timeline of preconditions leading up to the event, as well as a detailed description of the event.

23. Corrective Actions: List all short-term and long-term corrective actions that were/will be taken to prevent recurrence of the incident. Short-term corrective actions are those actions taken that will allow return to work in short time frame. Long-term actions are more ‘programmatic’ in nature and typically include process revision, changes in training, ‘mistake proofing’, etc.

Note: Forms should be marked in accordance with the activity’s security and marking policies.

Figure 12-2 (2 of 2)

CRANE AND RIGGING ACCIDENT REPORT INSTRUCTIONS

1. Accident Category: Indicate either crane accident or rigging accident. Indicate if significant (see paragraph 12.3).
2. Reporting Activity/UIC: The activity and unit identification code responsible for reporting the accident in accordance with paragraph 12.6.2.
3. Activity Responsible for the Accident/UIC: Same as #2 above, or for NAVFACENGSYSCOM, provide FEC level UIC.
4. Report No.: The activity assigned accident number (e.g., Activity UIC-FY-CA-01).
5. Accident Location UIC: The activity and unit identification code of where the event took place.
6. Specific Location: The detailed location where the event took place (e.g., building 213, drydock 5).
7. Accident Date: The date the accident occurred.
8. Time: The time (24-hour clock) the accident occurred (e.g., 1300).
9. Contractor Operation: Check yes or no. If yes, enter contract number.
10. BOS Contractor: Check yes or no.
11. BOS Contractor equipment: Check yes or no.
12. Crane No.: The activity assigned local crane number (e.g., PC5), if applicable.
13. Crane Type: The type of crane involved in the accident (e.g., mobile, bridge), if applicable.
14. Category: Identify category of crane (i.e., 1, 2, or 3), if applicable.
15. Crane OEM: The original equipment manufacturer of the crane (e.g., Samsung, Grove, P&H), if applicable.
16. Crane Capacity: The certified capacity of the crane (e.g., 120,000 pounds), if applicable.
17. Hoist Capacity: The capacity of the hoist involved in the accident at the max radius of the operation, if applicable.
18. Weight of Load on Hook: The weight of the load on the hook, if applicable.
19. Weather: The weather conditions at time of accident (e.g., wind, rain, cold).
20. Critical lift: Was the crane or rigging gear being used in a critical lift? Check yes or no.
21. Ordnance Lift: Was the crane or rigging gear being used in a lift governed by NAVSEA OP-5? Check yes or no.
22. Lost Workdays? Check yes or no.
23. Fatality or Permanent Disability: Check yes or no.
24. Material/Property Cost Estimate: Estimate total cost of damage resulting from the accident. (See OPNAV M-5102.1).
25. Accident Type: Check all that apply.
26. Cause of Accident: Check all that apply.
27. Responsibility: Check all that apply.
28. Crane Function: Check all functions in operation at time of accident. Check N/A if a rigging gear accident.
29. Is this a recurring problem? Check yes or no. If yes, list Accident Report numbers.
30. Preparer: Printed name must be provided.
31. Concurrences: Printed name must be provided.
32. Certifying Official (Crane Accidents Only): Printed name must be provided.
33. WHE Program Manager/Contracting Officer: Printed name must be provided when.
34. Contracting Officers representative: Printed name must be provided for contractor crane or rigging accidents.

Enclosure (1)

Brief Description: No more than one paragraph summarizing the resultant incident.

Root Cause and Detailed Description: Provide the relevant background in a descriptive timeline of preconditions leading up to the event, as well as a detailed description of the event.
Corrective Actions: List all short-term and long-term corrective actions that are taken to prevent recurrence of the incident. Short-term corrective actions are those actions taken that will allow return to work in short time frame. Long-term actions are more ‘programmatic’ in nature and typically include process revision, changes in training, ‘mistake proofing’, etc.

Note: Forms should be marked in accordance with the activity’s security and marking policies.

Figure 12-1 (2 of 2)

CRANE AND RIGGING ACCIDENT REPORT

Accident Category: [ ] Crane Accident [ ] Rigging Accident [ ] *Significant Accident

Name and UIC of Reporting Activity: Government Contracting Official	**Copy To: Navy Crane Center** **Bldg. 491 NNSY** **Portsmouth, VA 23709** **Fax: 757-967-3808**

Name and UIC of Activity Responsible for the Accident: 0	**Activity Name and UIC of Accident Location:**	**Report No:**
	Specific Location:	**Accident Date:** / **Time:**
Contractor Operation: [x] Yes [ ] No **If Yes, Contract No:**	**BOS Contractor** [ ] Yes [x] No	**BOS Contractor Equip.** [ ] Yes [ ] No

Crane No:	**Crane Type:**	**Category:** CAT 1	**Crane OEM:**
Crane Capacity: Ton	**Hoist Capacity:** lbs.	**Weight of Load on hook:**	**Weather:** Clear no wind

Critical Lift/Critical Non-Crane Rigging Operation? [ ] Yes [ ] No

Ordnance Lift? [ ] Yes [x] No

Lost Work Days? [ ] Yes [x] No

Fatality or Permanent Disability? [ ] Yes [x] No

Material/ Property Cost Estimate:

Accident Type (check all that apply):

[ ] **Personal Injury*** [ ] **Overload*** [ ] **Two Blocked*** [ ] **Power Line Contact***

[ ] **Dropped Load*** [ ] **Derail*** [ ] **Overturned Crane*** [ ] **Crane Collision**

[ ] **Damaged Crane** [ ] **Damaged Load** [ ] **Load Collision** [ ] **Damaged Rigging Gear**

[ ] **Lower Threshold Crane Accident** [ ] **Other: Specify _____________** *** signifies significant accident**

Cause of Accident (check all that apply):

[ ] **Improper Operation** [ ] **Equipment Failure** [ ] **Inadequate Visibility**

[ ] **Improper Rigging** [ ] **Switch Alignment** [ ] **Inadequate Communication**

[ ] **Track Condition** [ ] **Procedural Failure** [ ] **Other: Specify** _____________

Responsibility (check all that apply):

[ ] **Crane Walker** [ ] **Rigger** [ ] **Rigger-in-Charge** [ ] **Operator** [ ] **Signal Person**

[ ] **Maintenance** [ ] **Management/Supervision** [ ] **Other: Specify** _____________

Crane Function:

[ ] **Travel** [ ] **Hoist** [ ] **Rotate** [ ] **Luffing** [ ] **Telescoping** [ ] **Other** [ ] **N/A**

Is this accident indicative of a recurring problem? [ ] **Yes** [ ] **No**

If yes, list Accident Report Nos.: _____________

ATTACH COMPLETE AND CONCISE SITUATION DESCRIPTION AND CORRECTIVE/PREVENTIVE ACTIONS TAKEN AS ENCLOSURE (1). Include root cause and contributing factors. Assess damages and define responsibility. For equipment malfunction or failure, include specific description of the component and the resulting effect or problem caused by the malfunction or failure. List immediate and long-term corrective/preventive actions assigned and respective codes.

INCLUDE: Printed Name, Code and Date.

Preparer:	**Phone:**	**E-mail:**	**Code:**	**Date:**
Concurrence		**Code:**	**Date:**	
WHE Program Manager or Contracting Officer/designee (if Applicable)		**Code:**	**Date:**	
Certifying Official (Crane Accident Only):				
Contracting Officer's Representative (if Applicable)				

Figure 12-1 (1 of 2)

Brief description:

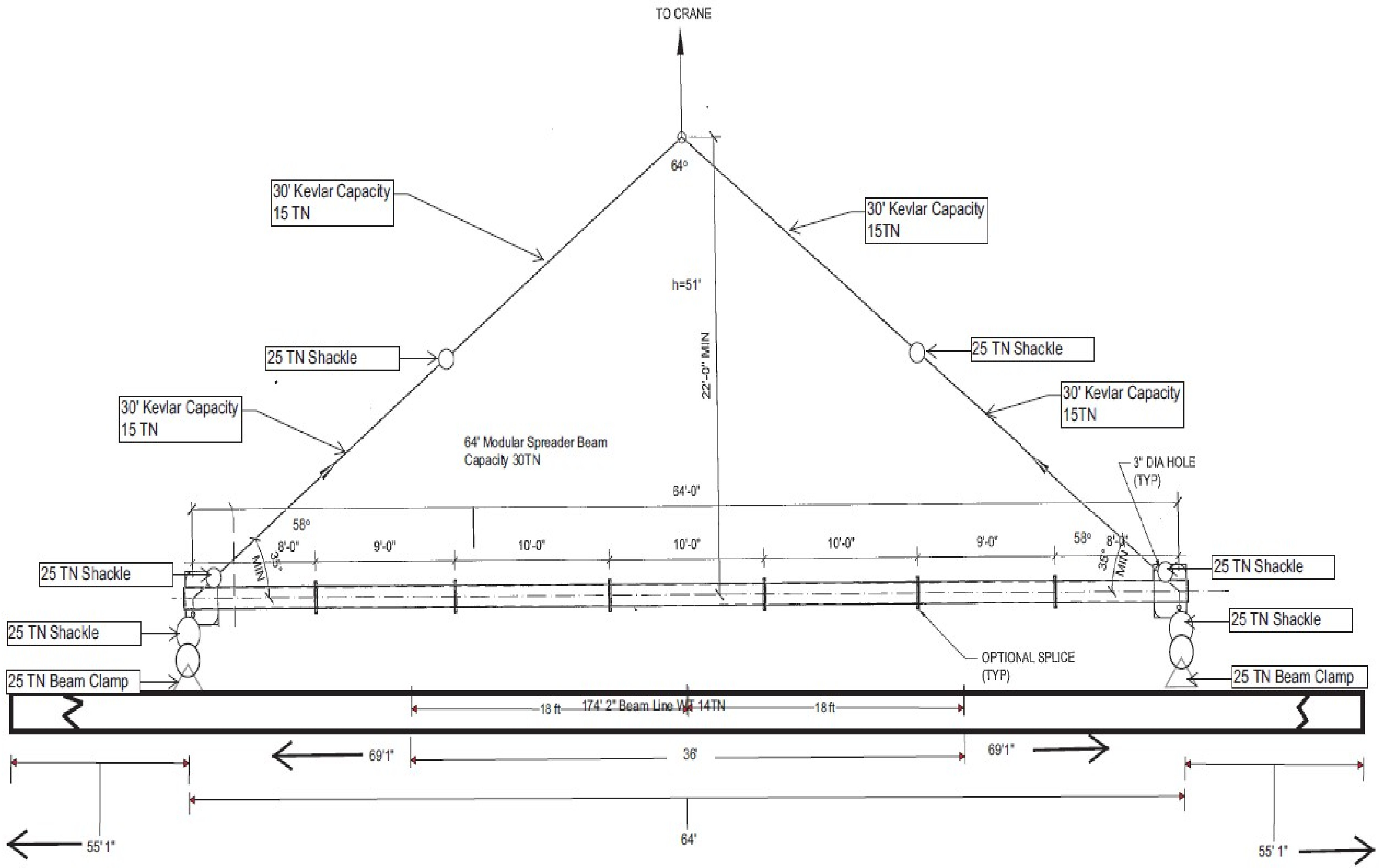
0

Root cause:

0

Corrective Actions:

Sample Rigging Diagram/Sketch



PROCEDURES FOR CONDUCTING MULTI-PURPOSE EQUIPMENT (MPE) LIFTS

1. Contractor to post Certificate of Compliance (P-1) in cab of MPE prior to entering job site.
2. NAVFAC P-2 Inspection will be conducted by a government contract Official / Representative prior to lift. Inspections will be maintained on file for one year by NAVFAC.
3. A government accepted Activity Hazard Analysis shall be available on-site and available for review by NAVFAC.
4. Contractor to provide equipment operator qualifications (ensure operator is trained to perform such lifts).
5. Contractor to provide rigger qualifications.
6. Contractor to ensure MPE OEM letter of authorization, load chart is posted in cab, and available for review by NAVFAC.
7. Contractor to ensure equipment operator’s daily inspection report is completed if required and made available for review by NAVFAC.
8. Contractor to notify NAVFAC CCO prior to the first lift or lifts over 20,000 LBS, lifts of hazardous material and complex / critical lifts.
9. Contractor to demonstrate equipment is properly configured prior to lift.
10. Contractor to provide proof from OEM that equipment is authorized for such lift, and lifts are made IAW OEM.

W.N RIGGING Equipment EXAMPLE

Activity Hazard Analysis (AHA)

Encl (1) - C410 KTR Crane Operations Requirement

AHA NUMBER/ NAME:		Overall Risk Assessment Code (RAC) - Use Highest RAC from Tasks Below					M
SERVICE ORDER NUMBER:		**Risk Assessment Code (RAC) Matrix**					
NAME & DATE ACCEPTED BY GDA:							
CONTRACT NUMBER:		**Severity**	**Probability**				
TASK ORDER #:							
PRIME CONTRACTOR:			Frequent	Likely	Occasional	Seldom	Unlikely
SUBCONTRACTOR:							
DATE OF PREPARATORY MEETING:		Catastrophic	E	E	H	H	M
DATE OF INITIAL INSPECTION:		Critical	E	H	H	M	L
COMPETENT PERSON:		Marginal	H	M	M	L	L
SITE SAFETY and HEALTH OFFICER		Negligible	M	L	L	L	L

RAC CHART and ACCEPTANCE BY GOVERNMENT DESIGNATED AUTHORITY	Review each **"Hazard"** with identified safety **"Controls"** and determine (RAC)
E = EXTREMELY HIGH	Identify the RAC (Probability/Severity) as E, H, M, or L for each "Hazard". Place the highest RAC at the top of AHA. This is the overall risk assessment code for this activity
H = HIGH RISK	**"Severity"** is the outcome/degree if an incident, near miss, or accident did occur and identified as: Catastrophic, Critical, Marginal, or Negligible after controls are in place
M = MODERATE RISK	**"Probability"** is the likelihood to cause an incident, near miss, or accident did occur and identified as: Frequent, Likely, Occasional, Seldom, or Unlikely after controls are put in place.
L = LOW RISK	
Prepared by (Name/Title/Date:	**Reviewed by SSHO/Site QC Manager (Printed Name/Signature):**
Project Safety Manager Review (Name/Date):	**Reviewed by Site Manager (Printed Name/Signature):**
	Reviewed by Subcontractor Representative (Printed Name/Signature):

Job Steps	Hazards	Controls	RAC
LHE Inspection and Operator Maintenance	• LHE not properly inspected prior to use • Manufacturer's pre-use inspection checklist or equivalent checklist not used to document inspection. • Dropped Loads Due to Rigging Failures	• Conduct and document a pre-use inspection of the LHE prior to use: • At a minimum, manufacturer's recommendations for a pre-use inspection must be followed. • In addition to the LHE, the hoist wire rope or chain and associated hardware (e.g., hooks and slings) shall be visually inspected by a qualified rigger. Rigger training credentials shall be provided with the Service Order request. • NEVER use any rigging that is not designed for the specific application. Example: Using a tow sling for lifting. • Note: In the event defects cannot be immediately resolved, notify your supervisor and remove the equipment from service by placing a red tag which states "OUT OF SERVICE" on the equipment • Submit the latest LHE inspection from the rental company along with the applicable lift plan, and the operator and rigger credentials/qualifications/ designation/training, etc., with the Service Order request.	L

		• Ensure deficiencies found during the pre-use or recent inspection have been corrected, e.g., repaired or replaced, before the LHE is placed into service. • Ensure only qualified personnel perform pre-use maintenance inspection as required. • Verify preventive maintenance has been performed prior to rental of the equipment. Failure to check will result in the contractor having to absorb the expensive of getting a serviceable replacement. • Verify the LHE Equipment Log is up-to-date and is maintained with the LHE.	
LHE Setup	• LHE failure due to improper setup and evaluation	**LHE Configuration:** • Evaluate ground conditions at lift location • Will the LHE be level within 1%? • Will a Load Moment Indicator be used? • Rigging locations should be identified in advance **Proper Rigging:** • Obtain and document specifications and inspection records for all rigging • Obtain and document riggers certifications • Identify communication method for Riggers to Operator **Environmental Considerations:** • Determine what wind speed will stop lift (20 mph or manufacturer recommendation?) • Ensure proper line of sight visibility is available for operator. Blind lifts are considered critical lifts. • Determine who will calculate additional weights if inclement weather brings rain/ice • Determine power line proximity **Lift personnel need to be identified and certs supplied:** • Lift Director • LHE Operator • Riggers • Signal Persons • Other/Supervisor • Pre-lift meeting needs to be held to ensure all personnel understand all aspects of lift. • All personnel not directly involved with lift should be out of swing radius/lift area during lift operations.	**L**
LHE Operation - Traveling, Traversing and Positioning for Use	• Collisions, road conditions, poor vehicle maintenance can result in equipment collisions, personal injury, or death • Contact with Overhead Obstructions • Contact with Overhead Utilities e.g., electrical lines • Personal Injury or Death • Property Damage	• Equipment travel will be conducted in and around the jobsite. • Cell phones shall not be used when operating the equipment. • A jobsite speed limit will be established by the SHO commensurate with conditions terrain and worksite conditions. • Equipment lights, tires/tracks, and dashboard lights shall be inspected prior to departing. • Maintain awareness of personnel on foot and other vehicles. • Always use a spotter when backing up equipment. • If equipment is not equipped with an audible backup alarm or becomes defective during use, at least one spotter and possibly two spotters may be necessary in order to warn workers to clear travel path and identify any potential hazards such as a fire hydrant or other utilities or equipment and structures. • Utilize spotters to verify no one is lying or leaning on the equipment, if ground hazards exist, in close proximity to structures, pipe racks, buildings, etc., and if backing is required. • Spotter/Ground Guides will remove debris and other identified obstacles and/or encumbrances will be marked or barricaded. • A Performance Class 2 or Class 3 High Visibility (Hi-Viz) apparel shall be worn based on site conditions.	**L**

		• NOTE: For the majority of work performed under the FUELS program, a Class 2 Hi-Viz Vest shall be worn by spotters. • Competent Person and equipment operator will walk the travel path of the excavator to make note of any potential hazards that may interfere with the safe movement and operation of the equipment. Take all necessary precautions to reduce the risk of damaging facilities, fuel systems and any interferences from Overhead Obstructions and Utilities. • Overhead hazards shall be evaluated prior to moving equipment on the project site. • Overhead power lines shall be shut-off and locked-out (if feasible), because shutting off power to some facilities may impact other mission requirements disassociated with the SO work task. • Areas with overhead hazards shall be barricaded with caution tape to prevent contact. In areas where it is not feasible to use barricades, spotters shall be provided: however, the minimum distances from electrical lines must be observed.	
Operations – Hand Signals	• Personnel Injury • Equipment Damage • Inability to Effectively Communicate with Signal Person	• A signal person must be used in the following situations: • When the point of operation, load travel, area near or at load placement, is not in full view of the operator. • When the equipment is traveling adjacent to fuel system piping, equipment, structures; and • The view in the direction of travel is obstructed. • Ensure a signal person is present when the operator does not have full view of the lifting operation. • Due to site-specific safety concerns, either the operator, lift supervisor, competent person or the person handling the load determines that it is necessary. • During LHE/LHE operations requiring signals, the ability to transmit signals between the operator and signal person shall be maintained. • If ability to communicate is interrupted at any time, the operator shall safely stop operations requiring signals until it is reestablished, and a proper signal is given and understood. • Only one person gives signals to an Operator at a time unless an emergency stop signal is given (which may be given by anyone and must be obeyed by the operator). • Signal Person must know and be able to convey the Standard hand signals to the LHE Operator. • LHE Operator shall only recognize signals from the signal person to guide and control movements of the equipment and load. • Ensure personnel acting as signal person during LHE operations is clearly identified to the LHE operator and personnel onsite, (e.g., different colored hard hat, high visibility garment - orange FRC vest, etc.). • Place a second signaler who can see both the primary signaler and the LHE operator, to relay signals, when the operator cannot see the primary signaler. • Discuss and agree on special signals in the pre-task tailgate for operations not covered by standard hand signals. Special signals will not conflict with standard hand signals. • Whether Manual or Mobile Device Signal are conveyed to Operator, the signals to be used shall be discussed during the pre-setup STEPS and/or prior to the lift. • Note: Radio communication will be on a different frequency than the Operating Area.	**L**

Inspection of Rigging	• Rigging failure • Cuts	• Wear work gloves • Rigging equipment shall be inspected as specified by the manufacturer, by a Competent Person (CP), before use on each shift and as necessary during its use to ensure that it is safe • The CP must have training and experience equivalent to, or be under the supervision of a Qualified Rigger (QR) as defined in EM 385-1-1 Appendix Q. **Make a thorough inspection of slings and attachments. Items to look for include:** ▪ Missing or illegible sling identification, ▪ Acid or caustic burns, ▪ Melting or charring of any part of the sling, ▪ Holes, tears, cuts, or snags, ▪ Broken or worn stitching in load bearing splices, ▪ Excessive abrasive wear, ▪ Knots in any part of the sling, ▪ Discoloration and brittle or stiff areas on any part of the sling, ▪ Pitted, corroded, cracked, bent, twisted, gouged, or broken fittings, and ▪ Other conditions that cause doubt as to continued use of a sling. • Where any such damage or deterioration is present, remove the sling or attachment from service immediately	**L**
Use of Tag Lines	• Personnel being struck-by, caught-in or, between the lift and loads. • Swinging overhead hazards	• Use tag lines to control the load • Wear hard hat • Wear hand and feet protection • Never stand underneath elevated loads	**L**
Communications	• Personnel being struck-by, caught-in or, between the LHE and loads	• Manual (hand) signals may be used when the distance between the operator and signal person is not more than 100 ft. (30.5 m) • Radio, telephone, or a visual and audible electrically-operated system shall be used when the distance between operator and signal person is more than 100 ft. or when they cannot see each other	**L**
Use of Engine Hoist /LHE Lifting	• Surface encumbrances • Struck by • Backed over Unqualified operator(s) and rigger(s)	• Post installation Certificate of Compliance for LHE and Rigging in cab and in office trailer as applicable • All surface encumbrances shall be moved or supported, as needed, to safeguard employees • Workers shall wear hard hats, high-visibility reflective safety vests • Equipment operators shall use a spotter when they have an obstructed view to the rear • All non-essential workers shall remain outside of the equipment's swing radius and the work zone • Essential personnel will remain in a location where they can be seen by the operator at all times • All equipment shall have operational back-up alarms • Written proof of qualifications of equipment operators, riggers, and others involved in the hoisting operations must be obtained and reviewed	**M**

LHE Operational Requirements – Fueling (NA for PRL-PND-611)	• Fire from static Discharge or ignition of vapors	• LHE shall be shut down before and during fueling operations. Closed systems, with an automatic shut-off that will prevent spillage if connections are broken, may be used to fuel diesel powered equipment left running. • One (minimum) dry chemical or CO2 fire extinguisher with a minimum rating of 10-B:C installed in the cab or at the machinery housing	L
(Space for field personnel to write-in additional site-specific steps, hazards, and controls)	•	•	

Equipment	PPE	Competent or Qualified Personnel Name(s)	Inspection	Training
• Personal Vehicles • Fire Extinguishers • First Aid Kits • Power Tools • Small hand tools • Heavy Trucks • LHE • Rigging • Cribbing • Engine Hoist	• Hard hats • Safety toed boots with ankle supports • Safety glasses • Safety vests (Class 2 Reflective Safety Vest) • Gloves (as needed) • Insulated gloves (as needed) • Hearing protection (as needed)	• Competent Person: • • Qualified Person: •	• Initial inspection of all equipment and tools to be used. • Daily inspection of equipment and tools. • Daily EHS documented inspections. • PPE inspections.	• OSHA 30-hour Construction course (SSHO minimum) • First aid/CPR/BBP • Initial site safety briefing • Daily tailgate safety briefings • Vehicle training • Proper license to operate heavy equipment • Fall Protection (potential for assembly of LHE) • Medical Clearance (if required)

AHA Signature Page

In accordance with EM 385 01.A.13 Contractor-Required AHA "Work will not begin until the AHA for the work activity has been accepted by the GDA" The AHA shall be reviewed and modified as necessary to address changing site conditions, operations or change of competent/qualified persons. By signing below I understand, agree to, and will conform to the site rules set forth in this AHA, my respective company's EHS Planning Documents (including amendments and attachments), and those controls agreed upon during any site-specific health and safety briefing(s).			
Name	**Signature**	**Date**	**Briefing By**

Add additional pages as needed for additional signatures.

WATERFRONT CRANE/VEHICLE USAGE REQUEST

Please provide the following information a minimum of **TEN WORKING** days prior to the intended crane/ vehicle use date. Form will be verified by the PW Dept. POC for the required information and submitted for a structural assessment as necessary to PHNSY / NAVFAC Engineers PWD Contractor Crane Oversight will forward you the results of the assessment.

Required Assessment Completion Date: NAVFAC CCOI	Lift Date and Duration:

Crane / Vehicle Information:			Wheel / Axel Information:			
Manufacturer/Model:			Axel #	# of Wheels	Distance to next Axel	Axel Weight
Gross Vehicle Weight (GVW):			1			
			2			
Outrigger pad Spacing:			3			
Side to Side	Front to back	Offset	4			
			5			
			6			

Lifting Orientation/setup: ☐ Parallel to Wharf Edge ☐ Perpendicular to Wharf Edge

Travel Path:

Lift / Load Information:
Location/Dock Marks:
Payload Weight:
Payload Type:
Boom Length/Reach & Angle Rotation:
Outrigger Load with maximum load and reach:
O1: O2: O3: O4:
Outrigger Load with no load, short boom up, counterweight rotated over outrigger:
O1: O2: O3: O4:

User / Requesting Authority, Crane Operator and Supervisor Activity:

NAVFAC Hi Contractor Crane Oversight / Crane Contractor Equipment Specialist

Point of Contact:

Code:	WHPP73	Cell:	Wes 808-478-1941
Email:	wes.m.naganuma.civ@us.navy.mil	Pager:	Work OFFICE. 808-471-0808
Phone:		Fax:	

APPROVED AS NOTED: ☐ **DISAPPROVED:** ☐

Operators / supervisors shall perform travel & lifting operations in accordance with posted signs & pier markings.

Place 6'-0" x 4'-0" x 4-1/2" rigid dunnage with steel plates and HSS 5 x 3 x 3/8" framing below all outrigger pads.

BE ADVISED THAT THIS REQUEST FOR CRANE/VEHICLE USAGE MAY NOT BE APPROVED. REQUESTING ACTIVITY SHOULD ASSURE THAT CONTINGENCY PLANS ARE AVAILABLE.

CONTRACTOR CRANE COMPLIANCE REFERENCE CHECKLIST

Prime Contractor:	**Subcontractor(s):**	**Date:**	
Crane Owner:	**Equipment Type:**	**Mfg.**	**Year/Model:**
Serial Number:	**Rated Capacity:**	**Quad Certification Date:**	**Annual Certification Date:**
Rigging Gear Owner:	**Location of Operations:**	**Inspection / Surveillance**	**Duration of Contract:**
Contracting Officer	**Phone:**	**Contract Number:**	

	YES	No	N/A	P-2
1. Is the crane certification valid? -- IAW 29 CFR 1919.71(a)-(a)(2), Cal OSHA Title 8 Article 99, 5021 (a) 1&2				
2. Are the load test (quad) and annual certification documents in the crane? - Cal OSHA Title 8 Article 99, 5021 (a) 2 & 5025				
3. **Is there a certificate of compliance posted on the crane, (or in the on-site office) with the current operator's name listed? --** NAVFAC P-307 Section 11				1
4. Is the operator qualified to operate this specific type crane? ASME B30.5, 5-3.1.2(a) & Cal OSHA title 8, Article 98, 5006.1 & CFR 1926.1427 (a)				
5. Is there a copy of the OEM operator's manual for the specific make and model crane? -- ASME B30.5, 5-1.1.3(b)				
6. Is there a load rating chart specific to the make and model of the crane? -- IAW 29 CFR 1910.180(c)(2), & ASME B30.5, 5-1.1.3 (a)(1) & CFR 1926.1433 (b)				
7. Is the hoist reeved according to OEM recommendations? -- IAW ASME B30.5, 5-1.1.3 (a)(5) & 29 & Cal OSHA Title 8, Article 93, 4923 (a)				
8. Are operating limits set for specific weather conditions? -- ASME B30.5, 5-3.1.3 (e) (8)				
9. Is the crane equipped with a boom angle indicator? IAW ASME B30.5, 5-1.9.1(c)				
10. Is the crane equipped with a load indicating device? -- & ASME B30.5, 5-1.9.9.2 (long shoring)& 29 CFR 1917.46(a)(1) Ca OSHA Title 8, Article 93 §4924)				
11. Is the device calibrated in accordance with OEM recommendations? -- IAW 29 CFR 1917.46(a)(1)(ii) – long shoring & ASME B30.5, 5-2.1.6 (b)				
12. Is the crane equipped with level indicators? **–** ASME B30.5, 5-1.9.11(d)				
13. Is the crane equipped with drum rotation indicators? -- ASME B30.5, 5-1.3.2(a)(5)				
	YES	**NO**	**N/A**	**P-2**

CONTRACTOR CRANE COMPLIANCE CHECKLIST

14. Is the crane equipped with anti-two block devices? -- ASME B30.5, 5-1.9.9.1 & 29 CFR 1926.1416(d)(3)(i)&(ii) & CFR 1917.45(j)(10) & Cal OSHA Title 8, Article 98, 5004(e)(3)				
15. Are operator aids (motion and load limiting devices) and other safety devices functioning properly? -- IAW ASME B30.5, 5-2.1.6 & 29 CFR 1910.180(d)(3)				
16. Is the hydraulic system functioning properly (without leaks)? --IAW ASME B30.5, 5-2.1.2(d)(h) & 29 CFR 1910.180(d)(3)(iv)				
17. Is the electrical system functioning properly? -- IAW ASME B30.5, 5-2.1.2(g) & 29 CFR 1910.180 (d)(3)(vii)				
18. Are hooks and latches inspected for deformation, chemical damage, cracks and wear? –IAW ASME B30.5, 5-2.1.2(e) & ASME B30.10, 10-2.2.1.3c				
19. Are audible signal devices (horns and warning devices, backup alarms) functioning? -- IAW ASME B30.5, 5-1.9.11(c) & ASME 5-3.3.7				
20. Are warning signs for electrical lines posted on the crane? - IAW ASME B30.5, 5-1.9.11(g)				
21. Is the wire rope in satisfactory condition? -- IAW ASME B30.5, 5-2.4.3(1)(b) & 29 CFR 1910.180(g)(1) & 29 CFR 1926.1413 a				
22. Does the minimum number of full wraps of wire remain on the hoist drum at all times? -- IAW, ASME B30.5, 5-1.3.2 (2)(a) & Cal OSHA Title 8, Article 93, 4929				
23. Is the crane used for long shoring / cargo transfer? If so is the crane certified? – 29 CFR 1918.66(a)(1) & 29 CFR 1919.71(a) & 29 CFR 1919.72\ (a) & 29 CFR1917.50 (c)(1)				

Floating Cranes / Barge Mounted Mobile Cranes

24. Is the crane properly secured to the barge? Are all rules of 29 CFR 1437 (construction) followed? Are the outriggers/stabilizers blocked or the crawlers traveling in a defined space as allowed by29 CFR 1926.1437? 29 CFR 1400.137 (n) to 1437(n)(7)(iii)				**34** **35**
25. Is the crane equipped with a load indicating device, a wind speed indicator and a list and trim indicator? **--** IAW NAVFAC P-307 Section 11 & OEM requirments				
26. Does the crane have a modified load chart based on calculated list and trim?—29 CFR 1926.1437 (m) (4)				
27. Are rules of 29 CFR 1915 (ship repair) and 29 CFR 1437 (construction being followed? 29 CFR 1915, 1915.115.111 to 1915.120 & 1926.1437				**36**

Multi-Purpose Machine-Forklift, Construction Equipment or Base Mounted Hoist

28. Is there proof from the OEM (or qualified PE) that the machine is approved for suspended load lifting. and is there a load chart on the machine? IAW 29 CFR 1910.178 (a)(4) – (a)(6)				**37**
29. Are personnel being lifted with a base mounted hoist, multi-purpose machine or construction equipment, if so, are requirements followed?				**39**

CONTRACTOR CRANE OPERATIONS CHECKLIST

	YES	NO	N/A	P-2
1. Does the operator know the weight of the load?-- IAW ASME B30.5, 5-3.2.1.1 (c) & CAL OSHA Title 8 Article 98 4999 (a) & 29CFR 1926.1417 (o)(3)				3
2. Are riggers and signal persons qualified by a third party evaluator? Or the employer's in house qualified evaluator _____________ 02 - 29 CFR 1926.1404(r)(1) & 1926.1428(a)(b)(c)				
3. Are ground conditions suitable for crane set up (Conditions means the ability of the ground to support the equipment, including slope, compaction and firmness)?—29 CFR 1926.1402 b				7
4. Are outriggers/stabilizers required? -- IAW, ASME B30.5, 5-3.2.1.5(h), & 29 CFR 1910.180 (h)(3)(ix) & CAL OSHA Title 8 Article 98 4994 (b)				5
5. If outriggers/stabilizers are required, are they extended fully, are the wheels off of the ground, is the setup in accordance with the OEM? – IAW ASME B30.5, 5-3.2.1.5(h) & CAL OSHA Title 8 Article 98 4999 (c) (1) & 29 CFR 1926.1404 (q) (1) (2)				6
6. Is the crane/machine rated for on rubber lifts by the OEM's load chart, is the machine certified for on rubber lifts? IAW ASME B30.5, 5-3.2.1.5 (m) & 29 CFR 1926.1417 (u) – (u)(2)(ii)				9
7. Is the crane/machine level and blocked, is it on firm ground and or adequate supporting materials provided? -- IAW ASME B30.5, 5-3.2.1.5(a)(1), & 29 CFR 1910.180, (h)(3)(i)(a) & 1910.180(c)(1)(ii)(b) & CAL OSHA Title 8 Article 98 4999 (c) (1) & 29 CFR 1926.1404(h)(2)				7
8. If blocking is required, are the entire surface of the outrigger pad supported and the blocking material of sufficient strength to support the outrigger pad? -- IAW 29 CFR 1910.180 (h)(3)(i)(a),(b),(c) & 29 CFR 1926.1404 (q) (5) & (h)(2)-(h)(3)				8
9. Is the crane/machine configuration good for the intended load? -- IAW ASME B30.5, 5-3.2.1(b) & 29 CFR 1910.180, (h)(1)(i) & 29CFR 1926.1417 (o)(3)				4
10. Are personnel and obstructions clear of the swing radius of the counter weight with accessible areas barricaded to prevent damage or injury? -- IAW ASME B30.5, 5-3.2.1.5(a)(4) & CAL OSHA Title 8 Article 98 4993.1 & 29CFR 1926.1424 (a)(2)(ii)				10
11. Is the hook centered over load to minimize swing?-- IAW November 30, 2014 EM 385, 5-3.2.1.5(b)(3) & CAL OSHA Title 8 Article 98 4999 (d) (3) & CAL OSHA Title 8 Article 98 4999 (d) (3)				11
12. Are the lift & swing paths clear of obstructions? -- IAW ASME B30.5, 5-3.2.1.5(b)(3) & 29 CFR 1910.180 (h)(3)(iii)(b) & CAL OSHA Title 8 Article 98 4999 (e)(2) & 29 CFR 1926.1417 (p)(q)				13
13. Are personnel kept from standing or passing under the load? -- IAW ASME B30.5, 5-3.2.1.4(b) & 29 CFR 1926.1425 (a)&(b)(1)(2)(3)				15
14. Are proper signals used and is the operator responding properly to the signal person? -- IAW ASME B30.5, 5.3.1.3 (c) & CAL OSHA Title 8 Article 98 5008 (b) & 29 CFR 1926.1422				16 17
15. Are radios used for blind lifts, have they been tested on site before operations to ensure a clear and effective signal is transmitted? IAW ASME B30.5, 5-3.3.2 & 29CFR 1026.1420 (a)				17
16. Is the load secure in the rigging and lifted a few inches to ensure its balance? -- CAL OSHA Title 8 Article 98 4999 (d) (2)				12

CONTRACTOR CRANE OPERATIONS CHECKLIST

	YES	NO	N/A	P-2
17. Are tag lines used to control the load? (If rotation of the load is hazardous) -- 29 CFR 1910.180,(h)((3)(xvi) & CAL OSHA Title 8 Article 98 4993 (b)				14
18. Does the operator avoid sudden acceleration or deceleration of the load? -- IAW, ASME B30.5, 5-3.2.1.5(c)(1) & 29 CFR 1910.180(h)(3)(iii)(a) & Cal OSHA Title 8 Article 98 4999 (f) (1)				22
19. Is the operator's attention directed toward the load while operating the crane? -- IAW & ASME B30.5, 5-3.1.3(a) & 29CFR 1926.1417(d)				16
20. Is the operator in control of the crane so as not to allow the load or other parts of the machine to contact any obstruction? -- IAW, ASME B30.5, 5-3.2.1.5 (c)(2) & CAL OSHA Title 8 Article 98 4999 (e) (2)				
21. Are personnel prohibited from riding the load? -- IAW ASME B30.5, 5-3.2.1.5 (r) & CAL OSHA Title 8 Article 98 4995				21
22. Is the crane transited to and from the job site correctly per OEM instructions are empty hooks lashed during travel to prevent swinging? -- IAW ASME B30.5, 5-3.2.1.5 (l)(3) & 29 CFR 1910.180 (h)(3)(xiii)(c) & CAL OSHA Title 8 Article 98 4991 (b)(2)				2 18
23. Is boom side loading limited to freely suspended loads (side loading prohibited)? -- IAW ASME B30.5, 5-3.2.1.5 (d) & 29 CFR 1910.180 (h)(3)(iv) & CAL OSHA Title 8 Article 98 4999 (f)				20
24. Are power line in vicinity (if yes) is a critical lift plan addressing the requirements of 29 CFR 1926.1407–1411 provided) -- & 29 CFR 1926.1407 - 1411				25
25. Is there a critical lift plan for lifts over 75% of capacity of the machine, for personnel lifts, multiple crane lifts, pick and carry on rubber etc.. If so is the plan understood, initialed and signed off by the lift participants?				23 24 26
26. Does the operator remain at the controls while the load is suspended? –- IAW 29 CFR 1910.180 (h)(4)(i) & CAL OSHA Title 8 Article 98 4999 (h) (2) & 29CFR 1926.1417(e)(1)				19
27. When the crane/machine is left unattended, is it in a safe condition? IAW ASME B30.9,5-3.1.3 (e) & CAL OSHA Title 8 Article 98 4999 (i) (2) & 29CFR 1926.(e)				27

CONTRACTOR CRANE RIGGING GEAR CHECKLIST

	YES	NO	N/A	P-2
1. Is rigging gear inspected prior to use? -- I IAW & 29 CFR 1926.251 (a)(1) & Cal OSHA Title 8 Article 101, 5043				**30**
2. Is the rigging gear undamaged, acceptable for use? I**s defective rigging removed from service?** -- IAW CFR 29 1926.251 (a)(6)				**28**
3. Is the rigging marked with its working load limit and is it used in accordance with the limit? – IAW 29 CFR 1926.251 (a)(2) & ASME B30.5, 5-3.2.1.3(b) & Cal OSHA Title 8 Article 101 5045 (a)				**32**
4. Are positive latching devices (or mousing) on crane or rigging hooks? -- IAW, 29 CFR 1017.45 (long shoring)				**33**
5. Does rigging (gear) hardware, hooks, shackles, rings, pad eyes etc. meet applicable ASME Standards (e.g.B30.9 slings, B30.10 hooks, B30.20 below the hook lifting device, B30.26 rigging hardware)- IAW, 1926.251 (a)(1) & 1926.251 (a)(4) & ASME B30.26, B30.20, B**30.10** & B30.9				**29**
6. Are custom designed grabs, hooks, clamps, or other lifting accessories marked to indicate the working load limit? -- IAW 29 CFR 1926.251 (a)(4) & Cal OSHA Title 8 Article 101 5049 (f) & ASME B30.20, 20-1.2.1(a)(b)(c)				
7. Are wire rope clips used correctly? -- IAW 29 CFR 1926.251 (c)(5) & ASME B30.5, 5-1.7.3 (b) & Cal OSHA Title 8 Article 101 5045 (d) (3) & ASME B30.26, 26.3.1-26.3.1.3				**29**
8. Is wire rope used for hoisting, one continuous piece without knots or splices? IAW, & 29 CFR 1926.251(c) (4) (ii) & ASME B30.9, 9-2.3.1 a (1)				**29**
9. Are the eyes in wire rope slings properly fabricated without the use of clips or knots? -- IAW, 29 CFR 1926.251(c) (4) (i) & Cal OSHA Title 8 Article 101 5045 (f) & ASME B30.9, 9-2.3.1 a-d				**29**
10. Is sling protection used to protect slings and equipment from damage due to abrasion and sharp corners? -- IAW 29 CFR 1926.251(c)(9) & Cal OSHA Title 8 Article 101 5042 (a) (7)				**31**
11. Is the minimum length of wire maintained between the splices, sleeves or end fittings? IAW 29 CFR 1926.251 (c)(13)(i) & Cal OSHA Title 8 Article 101 5045 (b) (1) & ASME B30.9, 9-2.3.2 (b)(c)				**29**
12. Is the minimum circumferential length (Diameter) maintained for wire rope endless slings? -- IAW 29 CFR 1926.251(c)(13)(i)-(iii) ASME B30.9, 9-2.3.2 (d) & Cal OSHA Title 8 Article 101 5045 (b) (3) & ASME B30.9, 9-2.3.2(d)				**29**
13. Are wire rope and synthetic slings marked with manufacture's name, type of material, and rated capacity for all three configurations? -- IAW & 29 CFR 1926.251 (e)(1) & ASME 30.9, 9-5.7.1				**32**
14. Is the removal criteria for synthetic slings followed? -- IAW 29 CFR 1910.184 & ASME B30.9, 9-5.9.4				**29**
15. Is the removal criteria for wire rope slings followed? -- IAW ASME B30.9 9-2.9.4 & 29 CFR 1910.184 (f)(5) & Cal OSHA Title 8 Article 101 5045 (e) (1-8)				**29**

CONTRACTOR CRANE RIGGING GEAR CHECKLIST

	YES	NO	N/A	P-2
16. Are eyebolts, eye nuts and swivel hoist rings properly marked, are they in satisfactory condition, and are they used correctly? -- IAW ASME B30.26, 26-2.5.1 & 26-2.9.4.2-, 26-2.9.4.4				29
17. Are alloy steel chains assembled from grade 80 or higher chain?–IAW November 30, 2016, EM 385 15.D.& ASME B30.9,9-1.3.1(a)				29
18. Are alloy steel chain slings properly marked to show name of manufacturer, grade or size, number of legs, rated loads for the hitches used and reach? - IAW ASME B30.9,9-1.7.1 (a-f) & 29 CFR 1910.184(e)(1)				29
19. Is the removal criteria for synthetic round slings followed?—IAW ASME B30.9, 9-6.9.4				29
20. Is the minimum D/d ratio maintained between the sling and the pin, shackle, hook, or ring?				

SIGNATURE:	**Date:**

COMMENTS

CONTRACTOR CRANE OR RIGGING OPERATION CHECKLIST

	Before The Lift			
		YES	NO	N/A
1	Is the Certificate of Compliance, P-1, in the operator's cab (or in the contractor's on-site office for rigging operations) with the current operator's **Name listed**?			
2	Is the crane/machine transited to and from the job site correctly? Are the OEM instructions for travel being followed?			
3	If this is a critical lift, is a lift plan provided and understood?			x
3	If overhead power lines are in the vicinity, is a critical lift plan provided Addressing the requirements of 29 CFR 1926.1407-1411?			x
5	Does the operator know the weight of the load to be lifted (including rigging)?			
6	Is the load to be lifted within the crane/machine manufacturer's rated capacity in its present configuration?			
7	Are outriggers/stabilizers required and, if so, are they properly deployed and down?			
8	If outrigger/stabilizers are used, and the wheels are not off the ground is this the correct setup in accordance with the OEM?			
9	Is the crane/machine level and on firm ground, or if the ground is not firm are adequate supporting materials provided?			
10	If supporting materials are provided, is the entire surface of the outrigger/stabilizer pad supported and is the supporting material of sufficient strength to safely support the loaded outrigger/stabilizer pad?			
11	If the activity/location has specific ground loading restrictions or rules on ground bearing pressures, are those rules being followed?			
12	If outriggers/stabilizers are not used, is the crane/machine rated for on-rubber lifts by the OEM's load chart?			
13	Is the swing radius of the crane counterweight clear of people and obstructions and are accessible areas within the swing area barricaded to prevent injury or damage?			
14	Is the lift and rotation path clear of obstructions?			
15	Does rigging gear meet applicable ASME or host nation standards (e.g., ASME B30.9 for slings, B30.10 for hooks, B30.26 for rigging hardware such as shackles, swivel hoist rings, and eyebolts, B30.20 for below-the-hook lifting devices)?			
16	Is sling protection used to protect slings (especially synthetic slings) and equipment from damage due to abrasion and sharp corners and edges?			
17	Is the rigging gear used in accordance with its working load limit? Is the working load limit marked on the rigging gear?			
18	Was the rigging gear inspected prior to use?			
19	Is rigging gear undamaged and acceptable for the application?			
20	Are positive latching devices (or "mousing") used on crane and rigging hooks?			
21	If a mobile crane is used on a barge, are all rules of 29 CFR 1926.1437(construction) being followed?			
22	If a mobile crane is used on a barge are the outriggers/stabilizers blocked or are the crawlers traveling in a defined space as allowed by 29 CFR 1926.1437 (construction)?			
23	For floating cranes, are rules of 29 CFR 1915 (ship repair) or 29 CFR 1926.1437 (construction) being followed?			
24	If a multi-purpose machine, forklift, or construction equipment is being used, is there proof from the OEM (or qualified PE) that the machine is approved for suspended load lifting and is there a load chart?			

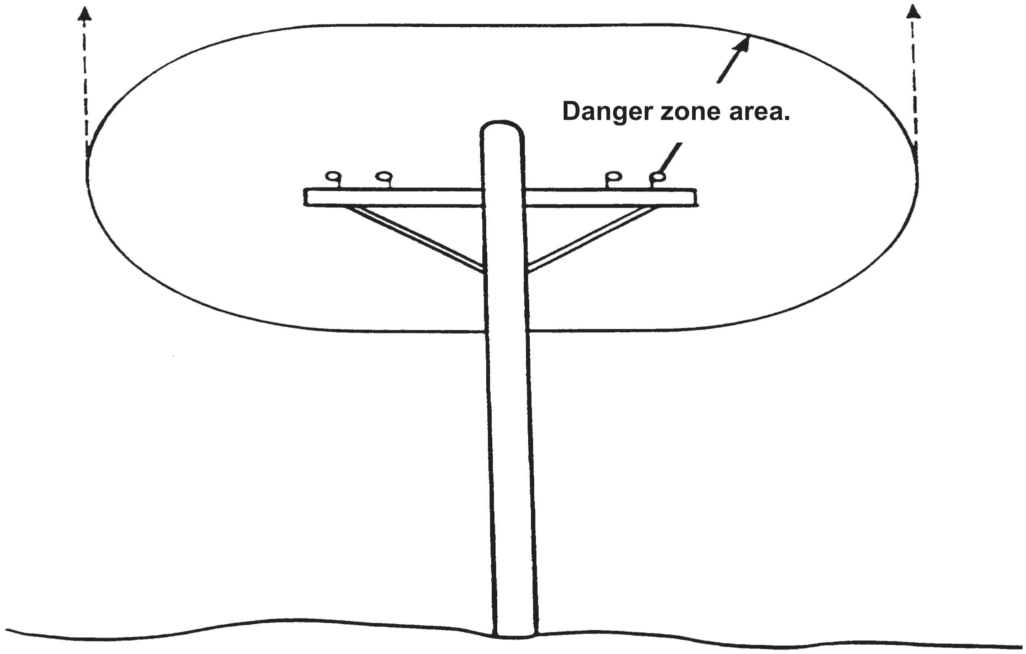
Figure P-2 (1 of 2)

		YES	NO	N/A
25	If a personnel lift is being performed with a crane or base mounted hoist, are all requirements of NAVFAC P-307, paragraph 11.1.g (5) for a crane or 11.1.m for a base mounted hoist being followed?			
	During The Lift			
26	Has the hook been centered over the load in such a manner to minimize swing and avoid side loading?			
27	Is the load well secured and balanced in the sling or lifting device after it is lifted a few inches for verification?			
28	Are taglines or other restraints being used, as necessary, for proper load control?			
29	Are personnel prevented from standing or passing under a suspended load?			
30	Are proper signals being used?			
31	Are radios being used for blind lifts?			
32	Is the operator paying full attention to and responding properly to the signals?			
33	Does the operator remain at the controls while the load is suspended?			
34	Is side loading occurring?			
35	Are personnel prevented from riding on a load?			
36	Are start and stop motions in a smooth fluid motion (no sudden acceleration or deceleration)?			
37	If pick and carry operations are allowed and performed, are OEM directions followed (e.g., rotation lock engaged, boom centered over front or rear)?			
38	When the crane/machine is left unattended, is it in a safe condition?			

Contractor:	Subcontractor:
Location:	Date:
Optional Information:	
Crane Operator:	
Make, Model, Serial Number of Crane:	
Load Weight:	
Crane Capacity at Pick Configuration:	
Rigging Equipment Capacity at Pick Configuration:	
Contract Number:	
Notes:	
Signature of Government Representative who completed this form:	

Figure P-2 (2 of 2)

DANGER ZONE FOR CRANES AND LIFTED LOADS OPERATING NEAR ELECTRICAL POWER LINES



Minimum clearance distances for operation near electric power lines and for transit with no load and boom or mast lowered.

VOLTAGE, KV (PHASE TO PHASE)	MINIMUM REQUIRED CLEARANCE, FT (M)
Operation Near High Voltage Power Lines	
0 to 50	20 (6.10)
Over 50 to 200	20 (6.10)
Over 200 to 350	20 (6.10)
Over 350 to 500	50 (15.24)
Over 500 to 750	50 (15.24)
Over 750 to 1000	50 (15.24)
In Transit with No Load and Boom or Mast Lowered	
0 to 0.75	4 (1.22)
Over 0.75 to 50	6 (1.83)
Over 50 to 345	10 (3.05)
Over 345 to 750	16 (4.87)
Over 750 to 1000	20 (6.10)

Note: For power lines over 1000kV the minimum clearance for operation and transit shall be established by the utility owner/operator or registered professional engineer who is a qualified person with respect to electrical power transmission and distribution but shall not be less than 50 feet and 20 feet respectively.

Figure 10-3